# Collective Trading Structure and Market Risk: From Aggregate Observations to Dynamic Inference

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Collective dynamics, inference and market risk

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**Abstract**

Can aggregate trading observations reveal collective structure that helps explain and assess market risk? We connect exact finite-network moments, effective trading weights and a fast–slow interdependent decision network to an observable-to-risk framework. Finite-network inversion and Bitcoin execution diagnostics distinguish recoverable statistical structure from investor identity. Common decision innovations increase conditional one-hour risk more strongly under weak simulated accommodation, yet more precise recovery of their average effect does not add forecast value beyond activity controls. Extending the same network to multiple factor inputs shows how genuine diversity buffers effective trading-scale expansion: at high scale, twelve independent group inputs reduce risk by 9.95% relative to a common input, while weak accommodation raises it by 57.40%. Neighbor influence and actual-return feedback amplify risk in the tested configuration; fast–slow coupling buffers it. A slow performance-allocation rule increases concentration without an established sixteen-hour risk increment. Official observed factor returns further show that signed covariance can hedge or amplify exposure risk, so factor labels alone do not measure diversity. Their historical prediction comparison does not establish a robust advantage over the supplied HAR references. Complete-state uncertainty and CDF-shaped calibration improve distributional inference under the model source, while direct Bitcoin interval transfer and frozen replay reveal the remaining market-calibration gap. The contribution is a quantitative account of how weighted dependence and its receiving environment transmit risk, with explicit tests separating explanatory mechanisms, observable inference and additional predictive value.

## 1 Introduction

Aggregate price and trading records compress many interacting decisions into returns, executed volume, counts and signed aggressive flow. A large decision can be executed in pieces; many independently acting traders can share a signal; and the same pressure can move prices differently as accommodation changes. The central question is whether a small observable interface can reveal risk-relevant collective structure, rather than merely reproduce a familiar market pattern.

The Cont–Bouchaud model links coordinated group actions to aggregate fluctuations[3]. Adaptive market models add switching and feedback[4], while calibration research distinguishes observable resemblance from parameter identification[8, 9]. We use these distinctions to connect a tractable finite-network reference to a weighted fast–slow decision model and an explicit observable-state inference task. A transaction price records an exchange under current information and trading constraints; our short-horizon model studies pressure and accommodation, not a universal intrinsic-value or eventual price-reversion law.

The research follows the sequence *macro observations → effective statistical structure → future joint trading/price paths → risk evaluation and application.* Exact moments and finite-ER inversion establish the initial inferential route. Weighted covariance and execution checks define what can be carried into market observations. The same interdependent network then supplies state dynamics, local perturbation response and common-signal/feedback experiments. Observable-only inference and common-path risk test the reverse route without supplying the target's true state or future.

Four questions organize the evidence: which structural quantities the reference identifies; which execution and temporal observations distinguish them; how the retained dynamic law turns state uncertainty into future risk; and whether that information improves an equally supplied forecast or reduces total measurement cost. Controlled simulations, Bitcoin observations and sector-portfolio applications have distinct roles. The completed experiments provide conditional mechanism results, numerical propagation checks and prediction comparisons; market observation calibration and independent predictive confirmation are evaluated as separate requirements.

A central inferential question is why a positive coordination effect may add little forecast value. Independent future repetitions separate genuine state heterogeneity from label noise. A paired capacity–correlation–feedback grid then quantifies the environments in which coordination raises risk. Finally, source-state information tests and a matched-field Bitcoin closure connect that explanation back to the observable interface. Their combined result is an environment-conditioned risk mechanism with a precise representation boundary: an interpretable state calculation is not automatically a uniquely informative forecast.

The strategy-ecology continuation then asks how growth in quantitative trading interacts with genuine input diversity. It changes group-factor covariance, effective order scale and accommodation within the same fast–slow network. A one-event Gaussian covariance argument provides the mathematical bridge; complete future paths test whether that bridge survives price transmission. Finally, a lagged performance-allocation rule makes exposure concentration dynamic. This separates three questions often conflated in discussions of quantitative trading: whether more strategy labels diversify inputs, whether scale exceeds the market's receiving capacity, and whether increasing concentration actually adds price risk at the measured horizon.

## 2 Finite network model and observable moments

### 2.1 Agents, actions, and the observation interface

Each period draws $H \sim G(N,p)$, where $p=\rho/(N-1)$, $N \geq 2$, and $0 \leq \rho \leq N-1$. Nodes are model traders, edges represent coordination, and a connected cluster $C$ contains $s_C$ traders. Each cluster shares $\xi_C \in \{-1,0,1\}$ with probabilities $q/2, 1-q, q/2$. Actions are independent across clusters and independent of the graph. Throughout, $q$ is the total activity probability. The implementation uses the equivalent per-direction probability $a=q/2$.

Each active trader submits a unit order $u>0$. A simplified external counterparty absorbs all submitted orders, and the log-price response is linear. The observations are

$$r=\frac{\lambda}{N}\sum_C s_C\xi_C, \qquad V=u\sum_C s_C|\xi_C|, \tag{1}$$
$$S_j=\sum_C s_C^j, \qquad \lambda>0, \quad 0<q<1.$$

Signed price effects can offset, whereas trading activity adds. Here $V$ counts executed activity and $\lambda$ is a model response scale. Table 1 defines the notation, and Fig. 1 illustrates the observation rule.

**Table 1.** Model notation. A connected group shares one action within a period.

| Symbol | Definition |
|---|---|
| $N, \rho$ | Number of traders and expected ER degree, with $p=\rho/(N-1)$ |
| $C, s_C, S_j$ | Connected group, its size, and the power sum $S_j=\sum_C s_C^j$ |
| $q, \xi_C$ | Total activity probability and shared sell/inactive/buy action |
| $u, \lambda$ | Unit order size and positive log-price response scale |
| $r, V$ | Log return and total executed model volume per period |
| $A_N(\rho)$ | Expected squared-size sum $\mathbb{E}S_2$ under the finite ER graph law |

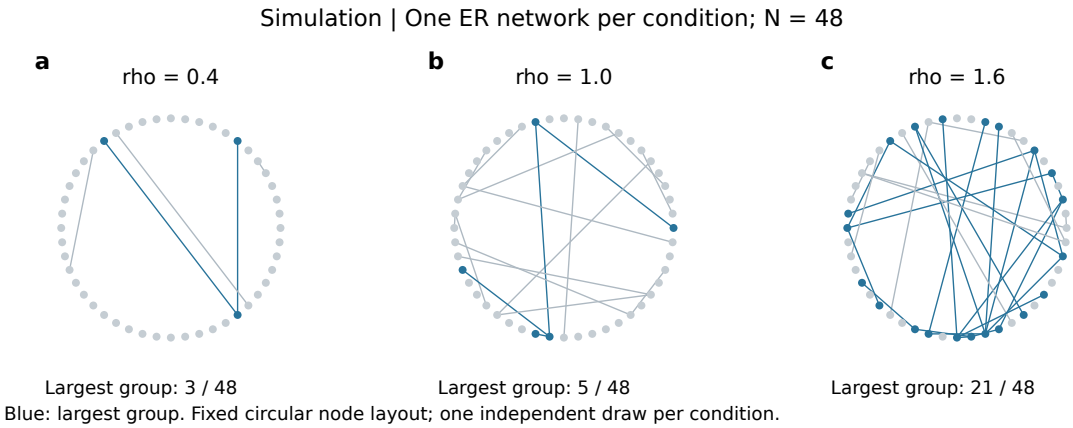



**Figure 1.** Shared group actions generate net price effects and total trading activity through the same execution rule. Nodes are model traders and connected groups coordinate their actions. [F01]

### 2.2 Exact moments and their mechanistic use

Zero-mean cluster actions eliminate cross terms in return variance. Conditional mean volume does not depend on the graph. Writing $A = \mathbb{E}S_2$ gives

$$\begin{aligned} \mathbb{E}V &= quN, \\ \mathrm{Var}(r) &= q(\lambda/N)^2 A, \\ \mathrm{Var}(V) &= q(1-q)u^2 A. \end{aligned} \tag{2}$$

Equal mean volume therefore need not imply equal risk. At fixed $N, q, u, \lambda$, a larger sum of squared cluster sizes increases return variance. For 100 independent traders, $S_2 = 100$; ten clusters of ten traders give $S_2 = 1000$. Variance differs tenfold in this model example while mean volume remains unchanged.

Higher-order relations retain the connection between collective activity and large price changes:

$$\begin{aligned} \mathrm{Cov}(r^2, V) &= u(\lambda/N)^2 q(1-q)\mathbb{E}S_3, \\ \kappa_3(V) &= u^3 q(1-q)(1-2q)\mathbb{E}S_3. \end{aligned} \tag{3}$$

Signed $r$ and $V$ have zero population covariance, while $|r|$ and $V$ are positively associated under this interface. Price direction and magnitude thus require separate analysis. The empirical analysis below tests stronger joint restrictions than this association alone.

Conditional on the graph, the joint characteristic function is

$$\mathbb{E}[e^{itr+izV} \mid H] = \prod_C [1 - q + qe^{izus_C} \cos(t\lambda s_C/N)]. \tag{4}$$

It depends only on the multiset of cluster sizes, directly supporting the identification boundary below. For fixed $N$, $|r| \le \lambda$, so the mathematical tail is bounded. This bounded support sets the domain for the finite-range tail tests below.

Source tables and methods: `docs/theory/FOUNDATIONS.md`.

## 3 Percolation and finite return distributions

In the infinite sparse ER limit, degrees are Poisson($\rho$) and $G_0(z) = G_1(z) = e^{\rho(z-1)}$. The edge-exploration generating function satisfies $H_1(z) = zG_1(H_1(z))$. A giant component emerges when the mean branching number crosses one. Its fraction satisfies

$$P_\infty = 1 - e^{-\rho P_\infty}, \qquad \rho_c = 1. \tag{5}$$

Below threshold, the expected cluster size of a uniformly selected node is $1/(1-\rho)$. In finite systems, $S_2/N$ differs from the statistic excluding one largest component,

$$\chi_f = \frac{\sum_{C \neq C_{\max}} s_C^2}{\sum_{C \neq C_{\max}} s_C} \tag{6}$$

which is recorded as missing when no finite component remains.

We scanned $N = 128, 256, 512$, nine $\rho$ values, and 40 graphs per cell, totaling 1,080 graphs. Figure 2 shows a smooth finite-size connectivity transition. Grid resolution and fluctuations affect peak positions. The infinite-system threshold organizes the finite-size comparison; the measured peak also depends on system size and grid resolution.

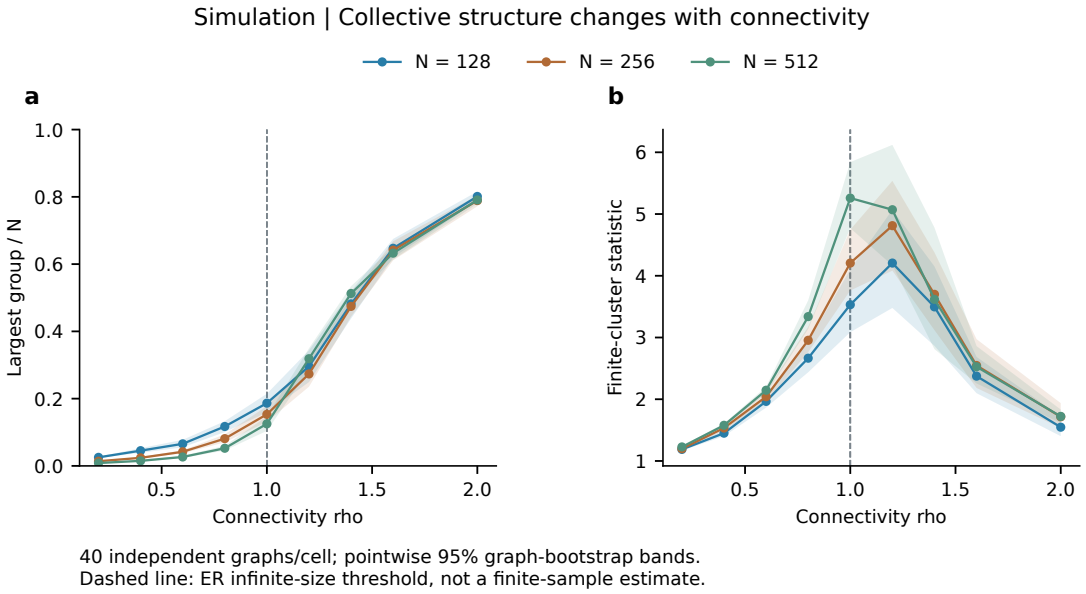



**Figure 2.** Finite-size connectivity transition for three population sizes. Each $N/\rho$ cell contains 40 independent graphs. The vertical line marks the infinite sparse-ER threshold; finite-system peaks depend on size and grid resolution. Source: simulation_study_v1/phase_summary.csv. [F03]

The mean-field critical law $n_s \propto s^{-5/2}$ concerns cluster number density. Selecting a node uniformly instead weights cluster sizes by $s$. Returns sum signed cluster actions; activity, cancellation, and bounded support affect their tails. For density $f(x) \propto x^{-\alpha}$, the complementary cumulative distribution (CCDF) exponent is $\alpha - 1$. These different sampling and aggregation operations require separate exponent definitions.

Five conditions each generated 8,192 periods, totaling 40,960 simulated intervals. Over the prespecified absolute net-order range $m = 4, \ldots, 128$, all three $\rho$ conditions rejected the simple finite-support pure power law. The raw $p = 0.005$ is the resolution floor of 199 parametric-bootstrap draws; Holm adjustment across three tests gives $p = 0.015$. A better relative AIC under some conditions does not imply adequate absolute fit. The implementation follows maximum-likelihood, Kolmogorov–Smirnov (KS), and alternative-distribution principles[6], adapted to finite integer support.

Collective amplification therefore survives even where this finite return distribution fails a pure-power-law fit. Cluster geometry and return tails answer different questions. Appendix B compares cluster distributions, population-size changes, and temporal aggregation.

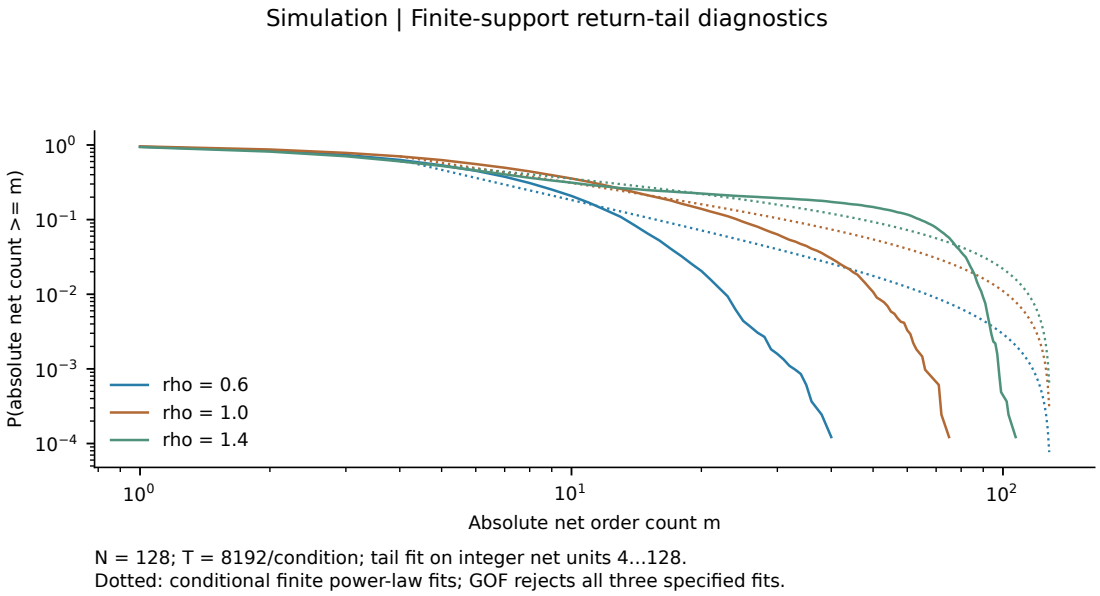



**Figure 3.** Return CCDFs and finite-support tail fits over the stated net-order range. The pure-power-law goodness-of-fit test rejects all three fitted connectivity conditions; relative AIC comparisons measure a separate property. [F05]

# 4 Parameter recovery and observational equivalence

## 4.1 Finite-graph moment inversion

For known $N, u$ under the fixed observation assumptions, population moments give

$$q = \frac{\mathbb{E}V}{uN}, \qquad A_N(\rho) = \frac{\mathrm{Var}(V)}{q(1-q)u^2}, \tag{7}$$
$$\lambda^2 = N^2u^2(1-q)\frac{\mathrm{Var}(r)}{\mathrm{Var}(V)}.$$

The ER mapping $A_N(\rho) = \mathbb{E}S_2$ is strictly increasing and therefore supports inversion. Let $C_s(p)$ be the probability that $G(s,p)$ is connected, and let $\pi_{N,s}$ be the probability that a uniformly selected trader belongs to a group of size $s$.

At fixed $p$, Gilbert's connectivity recursion is[7]:

$$C_s = 1 - \sum_{k=1}^{s-1} \binom{s-1}{k-1} C_k (1-p)^{k(s-k)},$$
$$\pi_{N,s} = \binom{N-1}{s-1} C_s (1-p)^{s(N-s)}, \tag{8}$$
$$A_N(\rho) = N \sum_s s\pi_{N,s}, \qquad C_1 = 1.$$

This classical recursion supplies the finite-size map needed by the moment estimator. The identity $A_N = N + N(N-1)\Pr(1 \leftrightarrow 2)$ connects it to pairwise connectivity. Coupling graphs by adding edges makes this probability strictly increasing in $p$ for $N \geq 2$, which gives the population inverse its uniqueness. The implementation used 380 decimal digits, checked against higher precision and small-graph enumeration.

Ten independently regenerated studies produced 4,920 windows of 256 periods, totaling 1,259,520 simulated intervals. Each study used 252 training, 84 validation, and 84 test windows, plus 72 unseen intermediate-parameter windows. Moment inversion and classical radial-basis-function (RBF) methods share evaluation units. The return-only view contains return standard deviation, mean absolute return, the 95th and 99th absolute-return quantiles, and the zero-return fraction. The volume-only view contains volume mean, standard deviation and 95th percentile. Joint inputs

combine these views and the Pearson correlation of volume with absolute return. Seeds and label metadata are excluded.

The supervised reference is kernel ridge regression with a Gaussian radial-basis kernel. For standardized observed features $x$, training labels $y_i = \rho_i$, and training mean $\bar{y}$,

$$K_{ij} = \exp[-\gamma \|x_i - x_j\|^2],$$
$$\widehat{\rho}(x) = \bar{y} + k(x)^\top (K + \alpha_K I)^{-1}(y - \bar{y}\mathbf{1}). \tag{9}$$

Here $k_i(x) = \exp[-\gamma\|x - x_i\|^2]$ and $\alpha_K$ is the ridge penalty. Median imputation, missingness indicators and standardization are fitted on training windows only. Validation MAE selects the penalty and kernel width from the fixed configuration grid. For positive return standard deviation, the scale-invariant view divides the three return-magnitude summaries by that scale and retains the zero fraction and volume summaries; degenerate windows follow the imputation rule. Table 2 and Fig. 4 report the matched-scale results.

**Table 2.** Connectivity recovery in the specified ER family: mean MAE over ten independent studies, in units of $\rho$.

| Method | Original grid | Unseen intermediate $\rho$ |
|---|---|---|
| Return-only RBF | 0.048660 | 0.045077 |
| Joint RBF | 0.027013 | 0.026882 |
| Volume-only RBF | 0.022196 | 0.036849 |
| Scale-invariant joint RBF | 0.032106 | 0.031013 |
| Moment inversion | 0.028372 | 0.027114 |

Each study regenerates training, validation, and evaluation windows. Source: depth_study_v1/summary_metrics.csv.

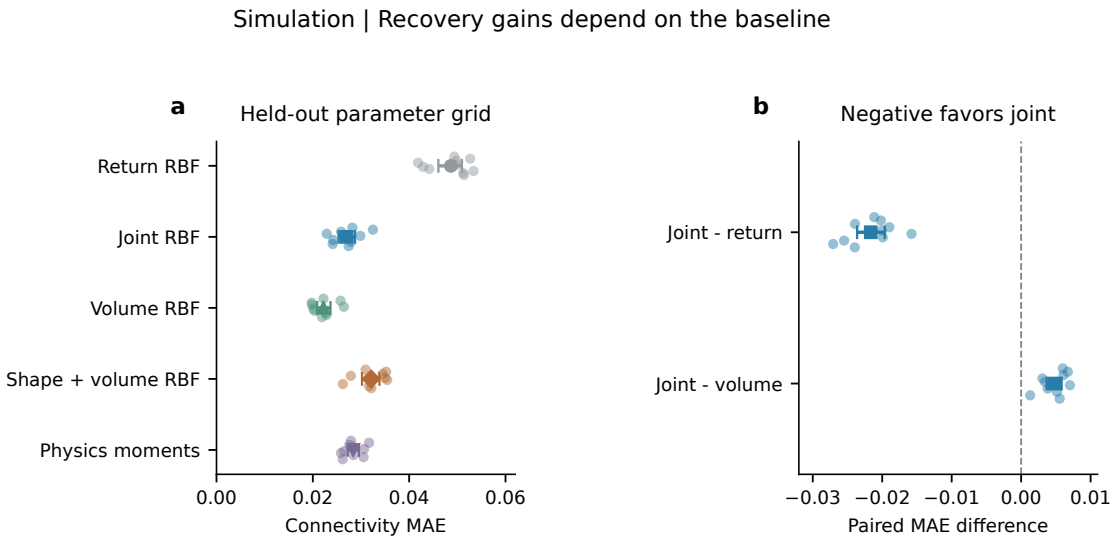



**Figure 4.** Parameter recovery across ten independently regenerated and retrained studies. Points are studies; summaries and intervals describe across-study variation. Table 2 gives grid and unseen-parameter MAE under the matched-scale design. [D01]

On the original grid, joint RBF reduced average MAE by 44.49% relative to return-only RBF. Volume-only RBF performed better than joint RBF in all ten studies. For unseen parameters, moment inversion achieved MAE 0.027114, compared with 0.026882 for joint RBF. Mechanistic formulas can therefore provide interpretable estimators within the specified family.

### 4.2 Price rescaling separates representation error from information loss

Each window undergoes $r \mapsto br$, with $b = 0.5, 1, 2$, while $V$ remains unchanged. These are paired transformations of the same windows. Doubling the price response increases raw joint-RBF MAE from 0.027013 to 0.258133. The scale-invariant representation and moment method remain at 0.032106 and 0.028372, respectively. Figure 5 shows this representation trade-off.

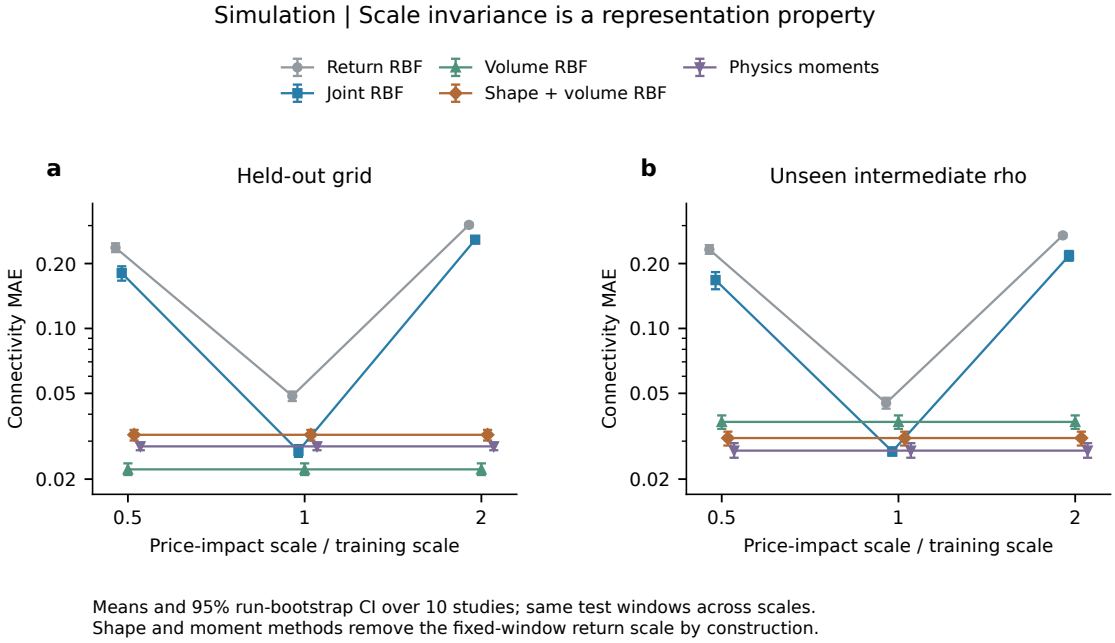



**Figure 5.** Price-response rescaling applied to the same evaluation windows in ten studies. Raw-return inputs change with scale, while moment inversion and scale-invariant inputs remove the specified window-level transformation by construction. Intervals resample studies. [D02]

### 4.3 Even complete observations need not reveal internal edges

The chain $\{12, 23, 34\}$ and star $\{12, 13, 14\}$ differ as four-node graphs, yet both form a single cluster of size four. Equation (4) gives identical complete joint price–volume laws. More passive observations of this kind cannot recover the internal edges. This is stronger than failure of a particular classifier. Exact certificates for general cluster mixtures are also included with proofs and rational checks.

Parameter recovery, structural equivalence, and information lost through feature compression require separate treatment. Financial ABM calibration studies likewise distinguish reproducing stylized facts from identifying mechanism parameters[8, 9]. The characteristic function, known-truth recovery experiments, and market restrictions thus address structural information, estimation accuracy, and model compatibility, respectively.

Source tables and methods: `docs/theory/DEPTH_ADDENDUM.md`.

## 5 Testing joint observations in Bitcoin

### 5.1 Data and price–volume comparison

The primary market case uses Binance Vision BTCUSDT spot five-minute bars[12]. The 2025 sample contains 105,120 bars and 365 complete UTC days, without filling or deletion. H1 provides calibration; H2 contains the original 184-day compatibility-test period. Returns are adjacent-close log changes, and bar volume is measured in BTC. Five-minute bars still compress the within-bar trade sequence. Figure 6 compares binned joint statistics and price-response proxies using total BTC trading volume.

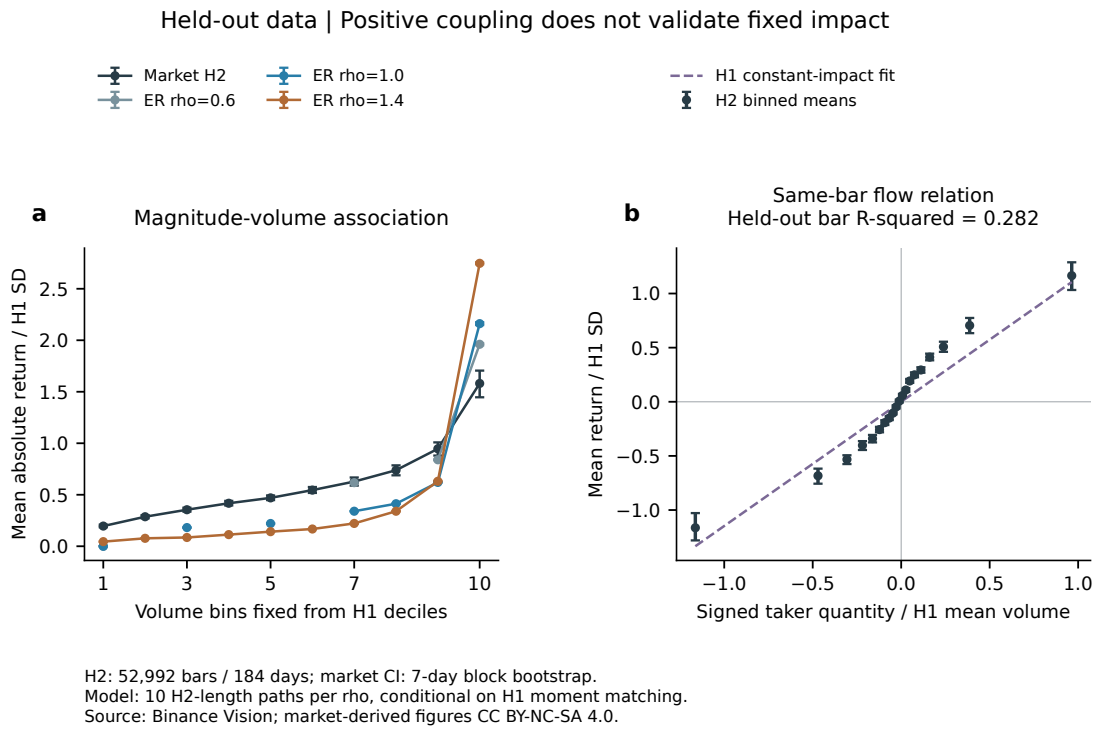



**Figure 6.** Bitcoin and model price–volume comparisons using H1-fixed bins. H2 magnitude–volume Spearman correlation is approximately 0.498, or 0.481 after seasonal adjustment. Empty model bins remain missing. Binance Vision, CC BY-NC-SA 4.0. [R01]

Appendix Fig. 32 compares market and model tails on a calibration-defined scale. The market bars record price and volume, while trader counts, clusters and $\rho$ remain unobserved. The empirical analysis therefore tests observable moment restrictions directly.

## 5.2 A joint necessary restriction derived from the mechanism

Eliminating scales from Eq. (3) gives a restriction for any graph distribution, beyond ER:

$$\Delta = \frac{1}{\mathbb{E}V}\left[\frac{\mathrm{Cov}(r^2, V)}{\mathrm{Var}(r)} - \frac{\kappa_3(V)}{\mathrm{Var}(V)}\right] = \frac{\mathbb{E}S_3}{N\mathbb{E}S_2} \in [1/N, 1]. \tag{10}$$

It requires fixed $N, q, u, \lambda$, symmetric independent actions, simplified execution, and linear response. Empirical calculations remove constant return drift and retain higher-moment and nonstationarity sensitivities.

For market H2, $\Delta = -8.181$, with an approximate 95% circular seven-day-block interval $[-11.998, -4.712]$. Figure 7 contrasts this estimate with the required positive range. The observed sign conflicts with the necessary range under the joint fixed-interface assumptions. All 12 calendar-month point estimates in 2025 are also negative, as are the checks excluding the highest-volume or highest-realized-variance day. These source-table sensitivities support using the restriction as a model diagnostic. The test identifies a failure of the fixed interface, while the next section examines how changing activity and response affects that restriction.

## 5.3 Temporal structure supplies additional forecasting information

Let $\mathcal{F}_t$ denote the information available through period $t$. With fixed parameters and independently redrawn graphs and actions,

$$\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = q(\lambda/N)^2\mathbb{E}S_2 \tag{11}$$

is constant. This static baseline does not itself provide a date-varying next-period risk forecast. Market H2 has lag-one magnitude autocorrelation 0.344, signed-return autocorrelation about

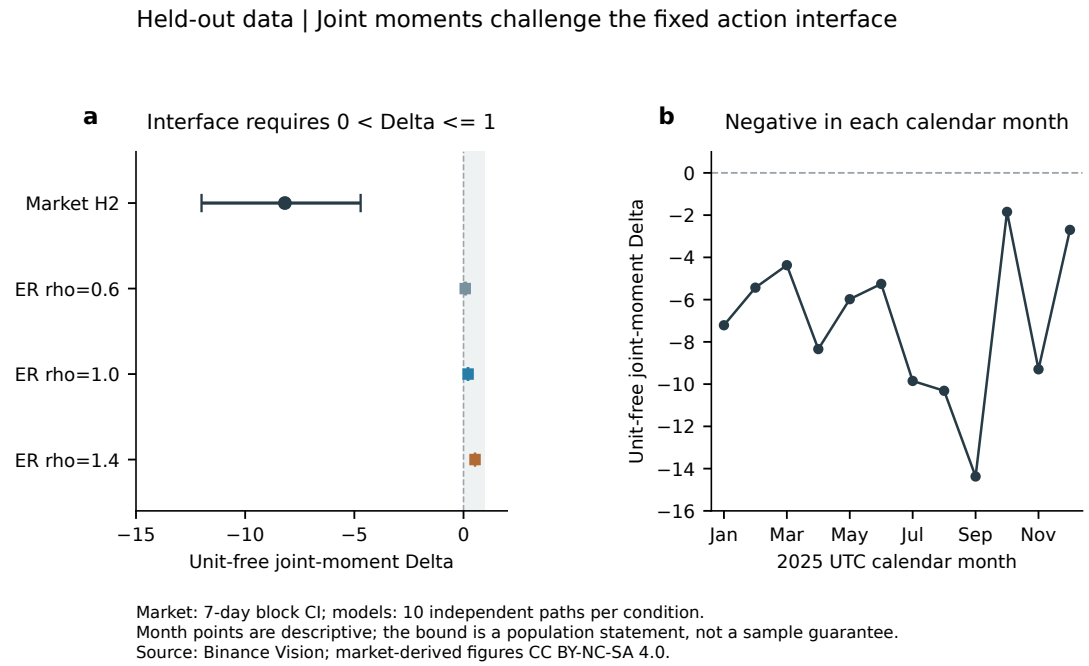



**Figure 7.** Joint-moment restriction for the fixed group-action interface and its BTC H2 estimate. The model requires a positive statistic. Market intervals use circular date-block resampling; monthly and extreme-day sensitivities are retained in the source tables. Binance Vision, CC BY-NC-SA 4.0. [R04]

−0.021, and a one-day aggregation variance ratio about 0.749. Temporal information therefore supplies an additional axis of mechanism comparison beyond the same-period cluster geometry.

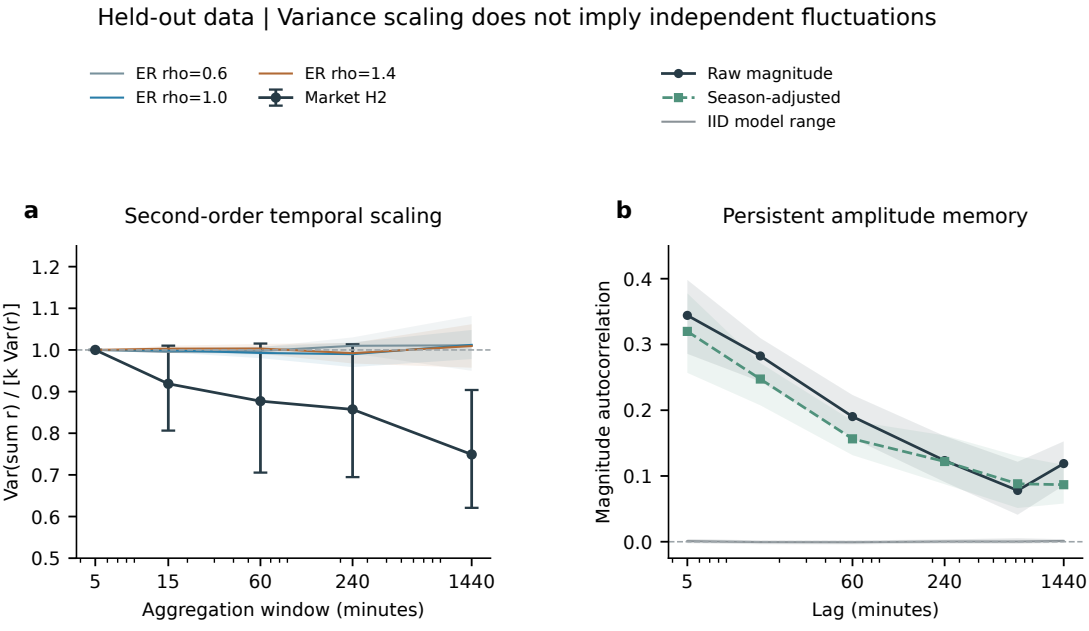



**Figure 8.** Temporal aggregation and lag dependence in independent model simulations and Bitcoin observations. Market magnitude memory contrasts with the independent-period baseline. Variance ratios and lag correlations describe temporal information, separately from cluster geometry. [R03]

# 6 Temporal states and forecasting information

To isolate temporal information, the controlled state extension keeps $N = 128, \rho = 1$ and independent graphs, while activity $q_t$ or response $\lambda_t$ follows an exogenous temporal state. Four arms each use ten paths of 8,192 periods, totaling 327,680 observations. Population mean volume and return second moments match across arms. Persistent activity produces magnitude and volume memory; persistent independent response can produce magnitude memory with nearly absent volume memory. Shuffling complete $(r, V)$ pairs preserves the same-period empirical joint distribution while removing temporal memory. Figure 9 separates these informational roles.

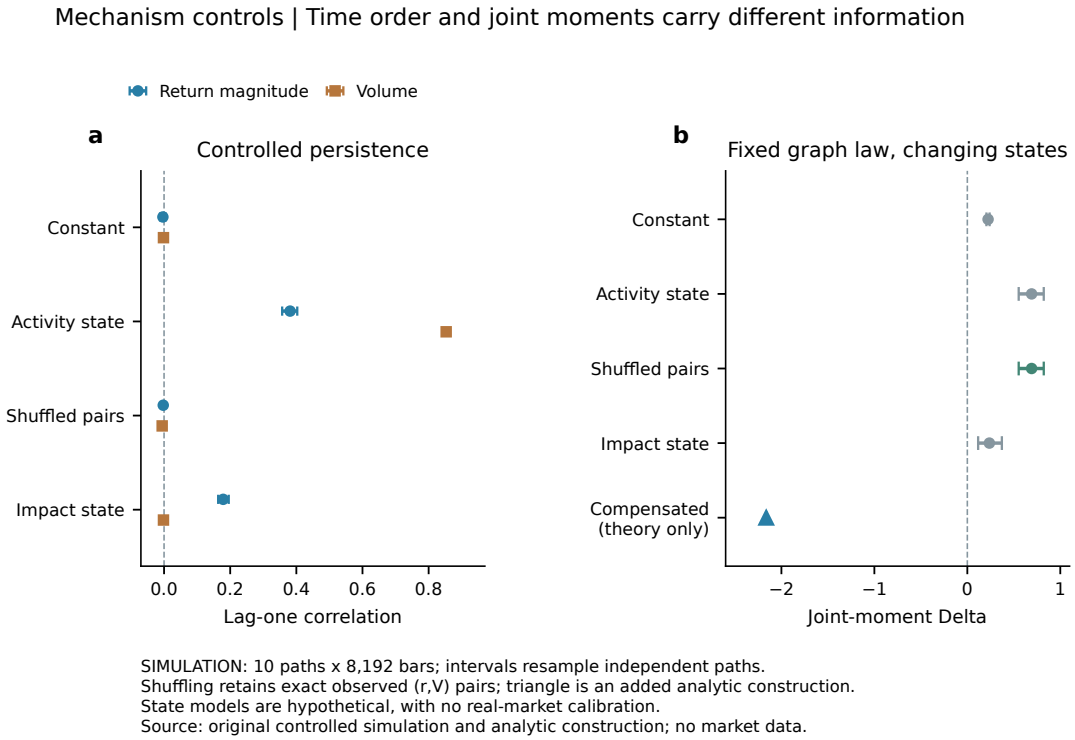



**Figure 9.** Controlled activity and response states under the same graph law. Persistent states produce different magnitude and volume memory; pair shuffling preserves the empirical same-period joint distribution and removes temporal order. Intervals resample independent paths. Sources: state_mechanism_v2. [V01]

Let $m_k = \mathbb{E}(q^k)$ and $A = \mathbb{E}S_2$, with activity independent of the graph. The total-variance identity gives

$$\mathrm{Var}(V) = u^2[(m_1 - m_2)A + N^2(m_2 - m_1^2)]. \tag{12}$$

The second term arises from changing activity, preventing interpretation of all volume variance as cluster structure. A complementary analytical construction, developed after the simulation comparison, sets $\lambda(q)^2 = \lambda_0^2 \bar{q}/q$, where $\bar{q} = m_1$ is mean activity. This keeps $\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = (\lambda_0/N)^2 \bar{q} A$ constant. Volume may reveal activity states without improving this forecasting target. Its population $\Delta \approx -2.16348$ shows that the positive fixed-interface restriction does not extend to all herding mechanisms. This construction demonstrates how activity–response coupling can reverse the restriction; estimating such a mechanism in market data remains a separate inference task. Full assumptions and exact checks appear in STATE_MECHANISM_v2.md.

An observable can identify the current activity state while its usefulness for forecasting return variance depends on state persistence and response compensation. This distinction motivates evaluating temporal observables in the forecasting benchmark.

# 7 Effective trading weights and the observable risk interface

The static group model supplies a tractable reference for a more general question: which statistical features of collective demand are needed for risk? Replace group size by a positive effective trading

weight $Q_C$. A large decision, several investors acting together, and one parent order executed in pieces can contribute similar aggregate pressure. Their participant interpretations differ, so effective weights are defined in units of trading quantity rather than people.

### 7.1 The covariance mechanism survives the extension

For action vector $\xi$ and fixed weight vector $Q$,

$$F = Q^{\mathsf{T}}\xi, \tag{13}$$

$$\begin{aligned}\operatorname{Var}(F \mid Q) &= Q^{\mathsf{T}}\Gamma Q \\ &= \sum_C Q_C^2 \operatorname{Var}(\xi_C \mid Q) + 2\sum_{C<D} Q_C Q_D \operatorname{Cov}(\xi_C, \xi_D \mid Q),\end{aligned} \tag{14}$$

where $\Gamma = \operatorname{Cov}(\xi \mid Q)$. The original independent symmetric-group law is the special case $\operatorname{Var}(F \mid Q) = q\sum_C Q_C^2$. With positive weights and marginal action variances fixed, additional positive cross-action covariance increases flow variance. Coupling can also change activity, mean direction, temporal dependence and price accommodation; its full price-risk effect is evaluated separately.

For execution sizes $v_i$ in one bar, define the independent-sign reference

$$B = \sum_i v_i^2, \qquad G_V = \frac{\operatorname{Var}(F)}{B}, \qquad G_E = \frac{\mathbb{E}F^2}{B}. \tag{15}$$

The averaging window and execution unit must be declared. $G_V$ is centered dispersion; $G_E$ also contains mean-flow energy. Neither is a measured fraction of followers. Under the fixed symmetric sign reference, $B$ is a size baseline; changing the grouping of executions changes that baseline.

A common-sign 100-unit parent split into ten 10-unit children has cumulative-flow variance 10,000 under a symmetric parent sign. Ten independent signs give 1,000. Both constructions have sum of squared child flows 1,000. In general,

$$\operatorname{Var}\left(\sum_t F_t\right) = \sum_t \operatorname{Var}(F_t) + 2\sum_{s<t} \operatorname{Cov}(F_s, F_t). \tag{16}$$

Thus directional persistence matters for cumulative risk, while it need not raise expected realized variance (RV) when per-bar second moments and a linear price response are fixed. This distinction determines the forecast target rather than serving as a cosmetic choice of risk statistic.

The execution bridge also yields a conditional inverse with heterogeneous individual sizes. Under independent symmetric group actions with common activity $q$, let i.i.d. positive individual marks have moments $\mu_1, \mu_2$, independently of the group structure and actions. Each active individual supplies one execution. For $\chi = \mathbb{E}S_2/N$,

$$G_E = 1 + \frac{\mu_1^2}{\mu_2}(\chi - 1), \qquad \chi = 1 + \frac{\mu_2}{\mu_1^2}(G_E - 1). \tag{17}$$

With equal marks, the amplification ratio equals group susceptibility. Exact finite enumeration and 300 independent simulation replications comprising 1.5 million periods verify the marked-size correction under its specified execution map. The map itself is identifying information: one participant splitting a two-unit parent into two unit fills and two participants sharing one common action can have the identical full $(F, V, K, r)$ event law. The inverse concerns an effective group statistic under the execution assumptions, not a universally recoverable head count.

### 7.2 What the public execution data actually constrain

The measurement audit covers 663 complete Bitcoin days (190,944 five-minute bars) and 24 dates with microscopic labels. Raw and exchange-aggregate records reconcile with bar totals. Three earlier fixed-date cases contain 12,536,460 raw executions and 4,378,760 aggregate units. Exchange aggregation is an observed execution grouping, not complete parent-order reconstruction.

Nonnegative sizes imply

$$\frac{V^2}{K} \le B_{\text{raw}} \le V^2. \tag{18}$$

The mean ratio of these arithmetic bounds is about 22,754 on the complete daily records. Macro $V, K$ alone therefore leave a broad size-concentration range. On sampled H2 dates, aggregate-execution $B$ is on average 2.391 times raw-execution $B$. A January-only Gamma size calibration gives inferred energy amplification 45.85 versus observed 41.71 in July, but 56.39 versus 25.80 in October. The latter comparison exposes transfer error in the size baseline.

For a declared execution unit $e$ with independent Gamma sizes of shape $a_s$, conditional normalized sizes are Dirichlet. Given its execution count $K^{(e)}$,

$$\mathbb{E}(B^{(e)} \mid V, K^{(e)}, a_s) = V^2 \frac{a_s + 1}{K^{(e)} a_s + 1}. \tag{19}$$

The working state rule targets exchange-aggregate $B$, predicting aggregate counts from the observed raw count through an H1-calibrated count map and using state-conditioned concentration. The fitted Gamma shape is a prediction parameter, not evidence that empirical sizes are i.i.d. Gamma. Relative to a refitted pooled-Gamma control, StateGamma reduces aggregate-$B$ loss from .2691 to .2549 (5.28%). The 24-date audit reconciles approximately 88.1 million raw executions and 29.2 million aggregate units from 48 checksum-verified archives.

Across 12 held H2 microscopic-label dates spanning six months, mean daily absolute relative amplification error is 35.14%; providing observed aggregate counts reduces it to 18.18%, at higher information cost. Measured and inferred $G_E$ share a numerator, so these discrepancies evaluate the execution baseline, not independent flow-variance recovery. Centered and uncentered statistics are reported separately.

Large count dispersion also has alternatives to coordination. In 256 independent repetitions per null construction, heterogeneous-intensity Poisson processes without self-excitation produce high Fano ratios. Their representative values, about 2,802 and 7,091 for varying-intensity and mixed-state constructions, coexist with a real count ratio near 13,784. A near-critical value obtained by substituting count moments cannot be identified with a market branching ratio or follower fraction.

Evidence record: measurement audit W1 and the fixed-date execution and macro-estimator amendments.

### 7.3 The risk map and its predictive requirement

For the original equal-size interface, $Q_C = us_C$ gives $r = [\lambda/(Nu)]F$, so the linear response below recovers its price law. For information $\mathcal{I}_t$, write $m_F = \mathbb{E}(F \mid \mathcal{I}_t)$ and a local response $r = \alpha + \beta F + \eta$. If $\alpha, \beta$ are fixed conditional on $\mathcal{I}_t$, $\mathbb{E}(\eta \mid \mathcal{I}_t) = 0$, and $\mathrm{Cov}(F, \eta \mid \mathcal{I}_t) = 0$, then

$$h = \mathbb{E}(r^2 \mid \mathcal{I}_t) = (\alpha + \beta m_F)^2 + \beta^2 \overline{B} G_V + \sigma_\eta^2. \tag{20}$$

This connects the group covariance mechanism to a return second moment. A same-window product $\overline{B}G_V$ is exactly the flow variance used to define it. Forecast information must therefore come from future state evolution, a cheaper valid measurement route, or a tested restriction that improves an information-matched forecast. Re-labelling a measured variance does not establish that increment.

## 8 Interdependent decisions, perturbations and future distributions

### 8.1 A weighted fast–slow decision network

The dynamic extension retains 48 effective units with fixed positive normalized weights $w_i$, two fixed sparse graphs and binary fast/slow states $s_i^f, s_i^s$. The fast layer responds to current neighbor pressure; the slow layer supplies a more persistent decision state. Each unit also carries an adaptation trace $z_i$. Interconnected opinion processes motivate this layered construction[15]; its financial interpretation and numerical rules are stated and tested here.

With capital-weighted neighbor direction $u_i^h$, the fields are

$$H_i^f = J_f u_i^f + \kappa s_i^s + \rho_f \mu s_i^f - a_f \mu z_i + h_t, \tag{21}$$
$$H_i^s = J_s u_i^s + 0.55\kappa s_i^f + \rho_s \mu s_i^s - a_s \mu z_i + 0.25 h_{\text{exec},t}.$$

The next fast sign is sampled with probability $[1 + \tanh(1.5H_i^f)]/2$ of being positive. The slow state attempts an update with probability 0.14 per step, and

$$z_i' = (1-\epsilon) z_i + \epsilon s_i^{f,\text{new}}, \qquad \epsilon = 0.055. \tag{22}$$

Here $J_f, J_s$ are within-layer influence strengths, $\kappa$ is cross-layer coupling, $\rho_f, \rho_s$ are inertia coefficients and $a_f, a_s$ are adaptation coefficients; these layer coefficients are distinct from the ER degree $\rho$. The inertia term favors retaining a direction; the adaptation term can reduce the attractiveness of a persistent orientation. Opposing positions are signs within this model. Layer coupling is same-unit fast/slow influence, not a measured institution-to-institution dependency or a rewiring rule.

Independent activity draws $A_i$ with probability $a = 0.58$ generate background quantities $C_0 w_i A_i$, with $C_0 = 1200$. For an additional deterministic pulse $d$,

$$F = d + C_0 \sum_i w_i A_i s_i^{f,\text{new}}, \qquad V = |d| + C_0 \sum_i w_i A_i. \tag{23}$$

The model count $K = \sum_i A_i$ records active effective units. It is not the number of exchange executions. A positive accommodation state $L$ and source noise convert flow into return through the frozen source law,

$$r = \text{clip}\left(0.009 \tanh\left[\frac{0.34(F/1200)}{L}\right] + \eta, -0.018, 0.018\right). \tag{24}$$

Accommodation is a simulated response state, not an observed order book. This law allows both flow alignment and absorption to affect price fluctuations. It does not contain an estimated intrinsic-value process.

### 8.2 A perturbation can induce trading without increasing every risk target

Paired-seed interventions compare the same background innovations with and without a pulse. Across 128 paired seed units, a 24-unit buy pulse raises first-step flow by 33.817 on average, with 95% paired interval $[30.062, 37.987]$. Subtracting the direct pulse leaves an induced background response of about 9.817 units. Thus the model gives a concrete route from a large trading decision to additional aligned trading.

The 48-step RV difference has mean $-1.50 \times 10^{-7}$ and interval $[-4.94 \times 10^{-7}, 1.36 \times 10^{-7}]$. At eight fixed latent validation states, the same positive pulse raises next-step flow second moment at four states and lowers it at four: a buy pulse can partly offset an initially negative state. These results distinguish a local response from a universal volatility increase. Without a scheduled opposite parent program, 108 of 128 seeds cross between opposite alignment thresholds. Such random internal switching is a model outcome, while strategic early exit and an inevitable consensus-to-crash cycle remain separate hypotheses.

### 8.3 Exact one-step moments and distribution-preserving propagation

Let $t_i = \tanh[1.5 H_i(S, d)]$ for complete state $S$, with the common-input channel off. Independent transition and activity draws give

$$m_F(S, d) = d + C_0 a \sum_i w_i t_i, \tag{25}$$

$$v_F(S, d) = C_0^2 \sum_i w_i^2 (a - a^2 t_i^2). \tag{26}$$

When a common future input is present, these formulas are conditional on that input, which must subsequently be integrated. Exact conditional $F, V, K$ means and covariances provide the observation interface. One-step Monte Carlo agrees with the formulas, and a 144-dimensional conditional-mean Jacobian agrees with finite differences to a maximum error of about $1.62 \times 10^{-10}$.

Repeated substitution of the mean state is a different operation: in a nonlinear model, $\mathbb{E}[f(S)]$ need not equal $f(\mathbb{E}S)$. A pair of hidden arrangements can have equal current $F, V, K$ and aggregate fast/slow directions but different next-step mean flows, 412.47 and 566.37 in a retained example. Those current summaries are therefore insufficient for exact closure.

The operational solution propagates complete stochastic states and uses their joint future paths. A verification design uses eight new latent states and three pulse conditions, with 12,288 prediction paths, 98,304 independent reference paths and 196,608 source-engine one-step branches. For the unpulsed 48-step flow mean, RMSE normalized by background quantity is 0.00785 for stochastic propagation and 0.38842 for mean iteration (Fig. 10). These are numerical propagation errors against an independent simulation reference, not forecast losses on market data. The result supplies a tested multistep continuation of the original network law.

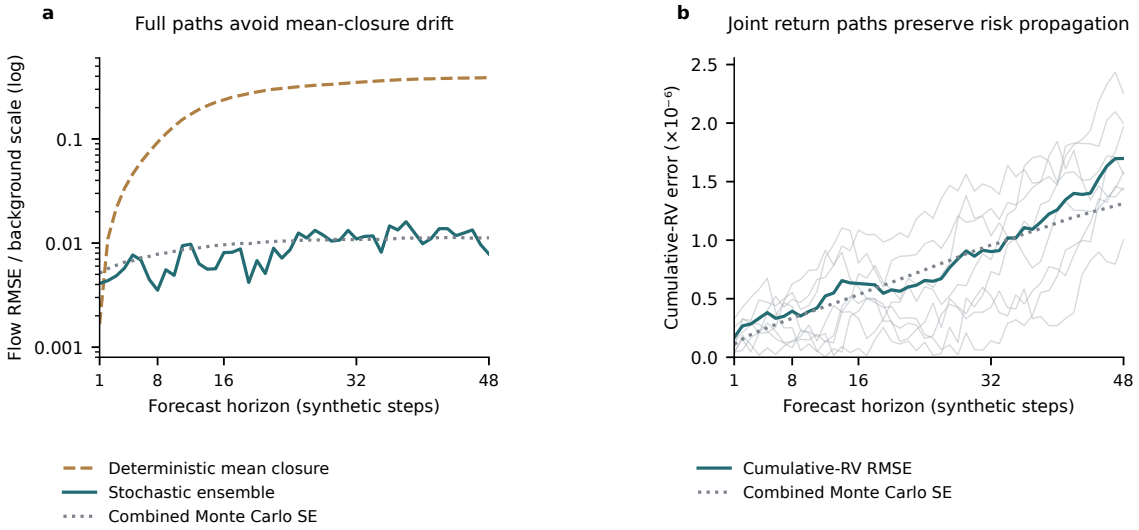



**Figure 10.** Full-state stochastic propagation and mean-state iteration against independent large-ensemble references. Eight fixed validation states, three pulse conditions and all declared horizons are retained. Monte Carlo agreement evaluates the propagation calculation, separately from latent-state inference and market forecast accuracy. Source: controlled continuation, D01.

Evidence record: network perturbation W2; exact mathematical interface and propagation D01.

## 9 Shared strategy signals and feedback on the same network

Similar inputs across trading strategies give an economic source of correlated decisions. Rather than attaching a separate predictor, we add a common signal to the retained decision fields and compare it with independent node signals having identical marginal distributions. Positive and negative lagged price feedback, together with a common activity rule, are tested as distinct channels.

The 2007 quant episode provides an economic motivation: factor-portfolio and transaction evidence is consistent with overlapping deleveraging and temporary withdrawal of market-making capital[18]. Funding/market-liquidity theory also supplies a conditional feedback channel between the ability to finance positions and the ability to absorb trades[19]. These precedents motivate separate shared-input and response mechanisms. Our experiment does not infer actual institution identities, reconstruct their portfolios or implement their complete margin rules.

### 9.1 A conditional link from shared information to covariance

For a shared standard Gaussian $Z$, positive loading $g = 0.15$ and the same fixed prior state, define $t_i(Z) = \tanh[1.5(H_i + gZ)]$. The independent-signal alternative substitutes independent $Z_i$. Node marginals and mean flow match, while

$$\begin{aligned}&\mathrm{Var}(F \mid S, \text{shared}) - \mathrm{Var}(F \mid S, \text{independent}) \\ &\quad = 2C_0^2 a^2 \sum_{i<j} w_i w_j \,\mathrm{Cov}_Z(t_i(Z), t_j(Z)) \geq 0. \end{aligned} \tag{27}$$

Both functions increase in the same scalar input. With an independent copy $Z'$, their covariance is one-half the expectation of $(t_i(Z) - t_i(Z'))(t_j(Z) - t_j(Z'))$, a nonnegative product. This standard covariance argument connects a shared strategy input directly to Eq. (14). Opposite loadings and changing activity or accommodation are outside this conditional inequality.

Numerical quadrature at all eight frozen validation states gives a one-step flow-variance increment of 24.1–35.5% relative to independent signals; 64- versus 128-node quadrature discrepancies are below $1.82 \times 10^{-11}$ in absolute variance units. Multiperiod return risk is then evaluated through the actual stochastic network rather than inferred from the one-step sign alone.

### 9.2 Paired economic-channel experiments

Six arms, including the baseline, generate 768 complete 240-step paths from 128 paired seed units; a further 64 baseline paths supply independent reference training. Table 3 reports all five interventions, with the ten RV/alignment comparisons retained under their frozen multiplicity rule.

**Table 3.** Controlled economic channels in one fixed network configuration. RV changes are percentages of baseline mean RV; alignment changes are percentage points.

| Channel | RV change | Alignment change |
|---|---|---|
| Common signal | +3.62% | +3.92 |
| Independent signals, same marginals | −1.78% | −0.52 |
| Positive lagged price feedback | +6.24% | +4.31 |
| Negative price feedback/accommodation | −3.99% | −2.74 |
| Common activity contraction | −6.20% | 0 exactly |

Independent-signal uncertainty intervals cross zero in the primary changes. The common activity rule reduces participation without altering the direction equation, so its alignment statistic is exactly unchanged. It is not a forced-liquidation or strategic-exit mechanism. The experiment identifies how specified shared inputs and feedback alter risk inside this network configuration; the corresponding causal attribution to real quant institutions would require additional evidence.

Evidence record: economic-channel design N01, all paired contrasts and common-input quadrature identity.

## 10 Observable-state inference and coherent future risk

### 10.1 From an observable prefix to a distribution of effective states

The inverse task now asks for risk-relevant effective state rather than every edge or participant. A student estimator receives past $F, V, K, r$; true weights, graphs, node states and future paths are withheld. The supplied 48-unit law and graph/weight distribution are explicit prior information. The first causal particle implementation uses 64 training, 32 selection and 64 new test trajectories. Future-data mutation leaves its predictions unchanged. Static-parameter resampling, however, reduces every tested graph/weight ancestry to one source, yielding zero-width HHI intervals. This implementation check motivates preserving multiple structural possibilities rather than interpreting a single remaining graph as identification.

A conditional simulation ensemble uses the same unchanged network law. Its bank contains 2,048 independent full trajectories, with separate 32-trajectory tuning, 64-trajectory calibration and 128-trajectory evaluation sets. Only 13 observable summaries of the past 32 bars are used

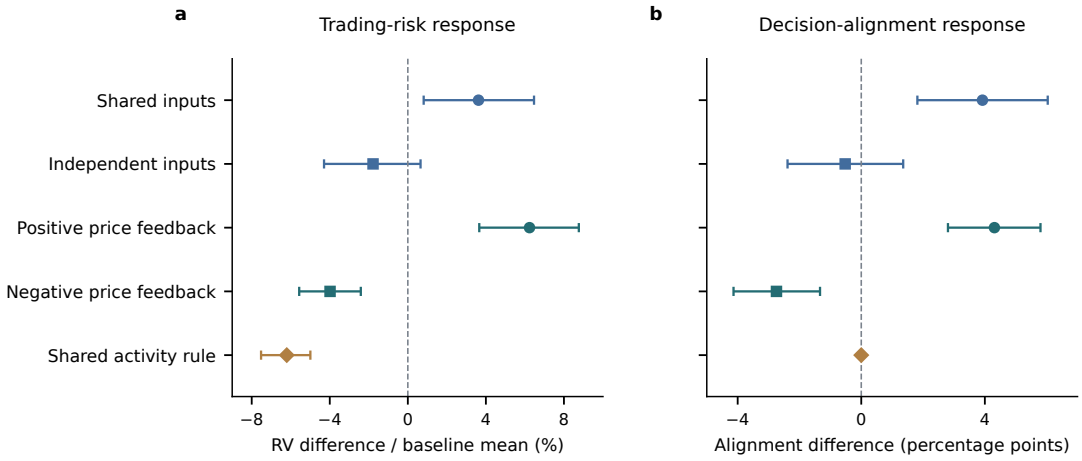



**Figure 11.** Paired shared-input, independent-input, feedback and activity interventions. The same seed supplies background innovations across arms. Both primary outcomes and every declared intervention are shown; the results concern this fixed source-law configuration.

to condition the bank. Standardization is fitted on the bank; neighborhood size is chosen among 32, 64 and 128 using tuning-set eight-step QLIKE, selecting 128. Every evaluated case retains 128 distinct source trajectories and their full future joint paths. This is approximate conditional simulation, not an exact Bayesian posterior.

For common nonnegative normalized scenario weights $\omega_j$, the next-step flow moments use total variance:

$$\bar{m}_F = \sum_j \omega_j m_F(S_j), \tag{28}$$
$$\mathrm{Var}(F \mid y) \approx \sum_j \omega_j v_F(S_j) + \sum_j \omega_j [m_F(S_j) - \bar{m}_F]^2.$$

The second term carries uncertainty about hidden state. Table 4 evaluates that state estimate on known-truth simulated units, not on unobserved market participant labels.

**Table 4.** Conditional simulation state estimates on 128 independent evaluation trajectories. The unconditional comparison uses the same simulation bank. Coverage concerns raw 90% state intervals.

| Target | Conditional RMSE | Prior RMSE | Coverage |
|---|---|---|---|
| Fast direction | .17567 | .39313 | 95.31% |
| Slow direction | .17015 | .42501 | 96.09% |
| Adaptation memory | .09030 | .28313 | 98.44% |
| Accommodation | .12295 | .15641 | 88.28% |
| Weight concentration (HHI) | .00477 | .00505 | 83.59% |

HHI intervals have mean width .00977, retaining uncertainty rather than a single repeated ancestor. Their 83.59% coverage is below nominal 90%; source diversity does not establish calibrated structural recovery. The simple $F/V$ fast-direction proxy has RMSE .13519, so the next inferential target is incremental information in slow state, memory, concentration and accommodation. The two inference designs use different evaluation samples and are not a same-sample before/after improvement experiment.

### 10.2 Generate return and flow moments from common paths

Network RV forecasts average squared returns generated by the nonlinear source law. A local decomposition $r = \alpha + \beta F + \eta$ checks their accounting without assuming residual orthogonality:

$$\begin{aligned}\mathbb{E}r^2 &= \alpha^2 + 2\alpha\beta\mathbb{E}F + \beta^2\mathbb{E}F^2 + \mathbb{E}\eta^2 \\ &\quad + 2\alpha\mathbb{E}\eta + 2\beta\mathbb{E}(F\eta).\end{aligned} \tag{29}$$

Common weights retain joint flow, accommodation, return and residual paths and preserve covariance positive semidefiniteness. All conditional-ensemble forecasts are positive without clipping. In contrast, fitting cross moments separately had produced 40 non-PSD cases in 1,192 selected rolling fits and 12 nonpositive calibrated risks. The coherent-path construction fixes that numerical inconsistency while leaving forecast accuracy to a separate test.

Expected $H$-step RV is $\sum_{h=1}^{H}\mathbb{E}(r_{t+h}^2 \mid \mathcal{I}_t)$. The second moment of a cumulative return additionally contains cross-time terms. These two outputs remain distinct throughout the experiments.

### 10.3 Separate predictive uncertainty from latent-state uncertainty

After freezing a positive point predictor $\widehat{h}(x)$, independent calibration units provide scores $s_i = |\log(y_i/\widehat{h}_i)|$. For nominal coverage $1-\alpha$,

$$\begin{aligned}k &= \lceil (n+1)(1-\alpha)\rceil, \quad q_\alpha = s_{(k)}, \\ C(x) &= [\widehat{h}(x)e^{-q_\alpha}, \widehat{h}(x)e^{q_\alpha}].\end{aligned} \tag{30}$$

For $k = n+1$ the order statistic is defined as infinity. With $n = 64$ and $\alpha = .1$, $k = 59$. This standard split-conformal construction[16] gives marginal coverage under exchangeability of independent calibration and evaluation units, conditional on earlier fitting/selection. It is not state-conditional coverage or a guarantee for dependent market dates. Risk-interval calibration does not calibrate HHI intervals.

A frozen 48-step particle-model evaluation uses 64 fresh calibration and 128 fresh evaluation trajectories. Raw intervals cover 118/128 in that new evaluation; calibration covers 121/128 with a larger mean width and slightly worse Winkler score. Its point QLIKE is .03594, compared with .03562 for the unconditioned source-law prior, with paired difference interval $[-.003365, .003888]$. Figure 12 retains this earlier particle-model test, separately from the conditional-ensemble comparison below.

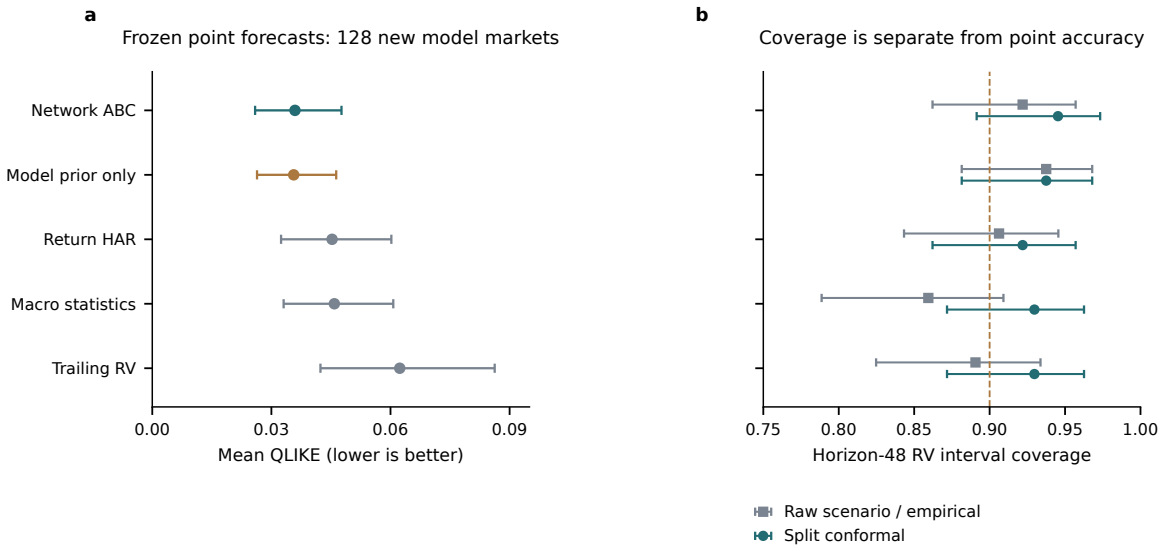



**Figure 12.** Observable-only inference and joint risk in the particle-model experiment. Source-law priors, test-state validation, interval width and calibration are displayed separately. This figure uses the first inference design and its frozen new calibration/evaluation sample; Table 5 uses a distinct conditional-ensemble design.

## 10.4 Risk comparison with equal simulation labels and macro features

The conditional ensemble is compared with a macro Gamma model using the identical 13 observable features and the same bank of future-risk labels. Fitted controls use the same bank and future-risk labels; the unconditional prior averages that bank, while trailing RV is a prefix-only baseline. Every method receives the same 64 interval-calibration labels. Eight steps is the fixed primary horizon, with one and 48 retained as secondary horizons.

**Table 5.** Primary eight-step risk on 128 independent controlled evaluation trajectories. QLIKE is smaller for better point forecasts. Coverage and width follow the shared 64-unit interval-calibration budget.

| Method | QLIKE | Coverage | Mean width |
|---|---|---|---|
| Conditional network ensemble | .13412 | 95.31% | $3.8 \times 10^{-5}$ |
| Same-feature macro Gamma | .13329 | 94.53% | $4.0 \times 10^{-5}$ |
| Unconditional network prior | .13688 | 95.31% | $4.1 \times 10^{-5}$ |
| Return-HAR | .13570 | 95.31% | $4.0 \times 10^{-5}$ |
| Trailing RV | .19185 | 94.53% | $4.9 \times 10^{-5}$ |

The paired QLIKE difference between the ensemble and Gamma is $+.000837$ (95% interval $[-.004832, .006098]$). Relative to the unconditional prior, the difference is $-.002756$ (interval $[-.015453, .008894]$). Incremental risk value over those two primary references is not established. Improvement over trailing RV survives the frozen four-comparison Holm rule, but that weaker reference does not replace the equal-feature and prior controls. At 48 steps, point scores .04304 and .04438 for ensemble and Gamma are reported as secondary outcomes, not a replacement primary horizon.

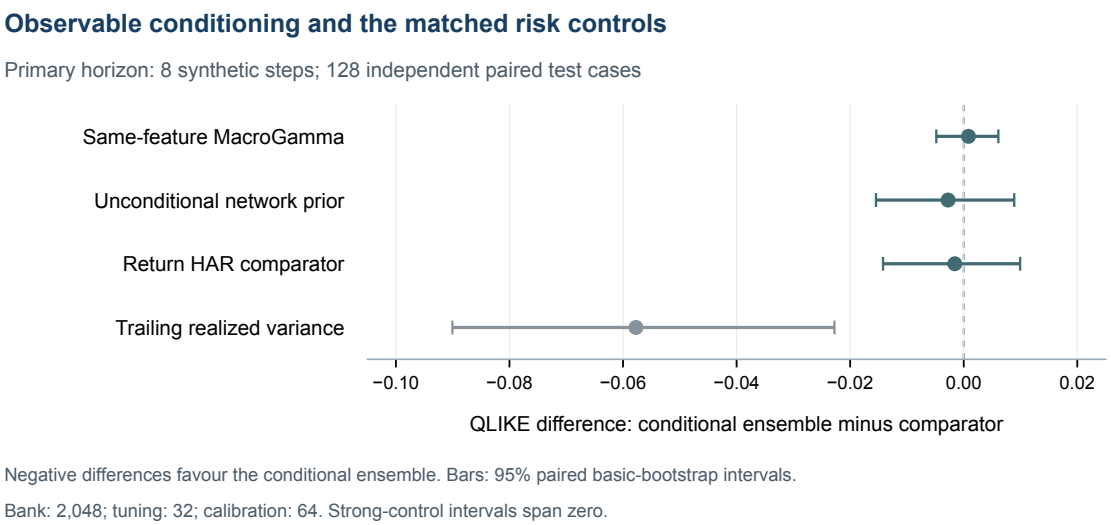



**Figure 13.** Primary eight-step conditional-ensemble risk contrasts: QLIKE difference from each prespecified reference, with 95% paired basic-bootstrap intervals; negative values favor the ensemble. The design uses 2,048 bank, 32 tuning, 64 calibration and 128 evaluation trajectories. The same-feature Gamma comparison shares observable summaries and simulated risk labels; the known generating law and offline labels remain disclosed costs. The three stronger-reference intervals cross zero. Trailing RV is a weaker comparator. Source: conditional-ensemble risk contrasts.

The full generative law, model age, unobserved initialization, offline state/risk labels, retained history and calibration are counted as information costs. A 32-bar deployed interface does not by itself establish savings over a statistical method using a longer observed history. The conditional-ensemble experiment establishes a working macro-to-state-to-risk calculation and a direct test of its added predictive value under disclosed simulation information.

Evidence record: observable inference I01, joint risk R01, calibration R02; conditional-ensemble state/risk scores and independent numerical recalculation.

### 10.5 Full-state uncertainty and distribution-shaped calibration

A calendar-time source study makes the observation clock explicit: eight network events form one five-minute bar, independent background executions supplement network trades, and half of predictable network impact is recovered. Section 11 develops this observation layer and its market diagnostics. For inference under that source law, complete particles are propagated: each particle retains its graph, weights, fast/slow signs, memory, accommodation, activity, clock and one of eight fixed bootstrap parameter instances. Source histories are drawn from the uniform eight-instance mixture, with 2,048 bank histories, 128 tuning histories, 256 calibration histories and 384 independent evaluation histories. The instance stays fixed along each history and continuation. This is a finite bootstrap-mixture source law, separate from the earlier nominal-law experiments above.

The distributional comparison allocates exactly 192 fresh futures to each method. The mixture samples 12 independent source blocks and 16 new futures per block; a legal MAP reference propagates the highest-weight complete source particle with the same $12 \times 16$ budget. Even a repeated source index receives new future innovations. Cluster-fair CRPS uses only pairs from distinct source blocks in its pairwise-distance term, preventing the 192 clustered continuations from being treated as independent draws from the mixture. Scores are normalized by the frozen bank's mean RV. On the 384 evaluation histories, mixture minus MAP has normalized fair-CRPS difference $-.022338$, with paired basic-bootstrap 95% interval $[-.040639, -.003961]$ and Holm-adjusted $p = .0378$ across the five declared primary comparisons. Retaining complete state and parameter uncertainty therefore improves this model's distributional prediction over premature selection of a single particle.

The interval construction also uses the shape of the same predictive distribution. If $G_x$ is its CDF, calibration scores $|G_x(y) - 1/2|$ give

$$C_{\mathrm{CDF}}(x) = \left[G_x^{-1}(1/2 - q),\ G_x^{-1}(1/2 + q)\right], \tag{31}$$

with the frozen finite-CDF tie/endpoint conventions, physical lower endpoint zero and upper endpoint infinity when required. Development data select the family before testing; 256 independent calibration histories set the 232nd order-statistic threshold. The alternative is the symmetric multiplicative log-ratio interval in Eq. (30).

**Table 6.** Distribution-shaped calibration on the same 384 source-mixture evaluation histories. Width and Winkler score have RV units; lower Winkler is better. These are model-source coverage results.

| Interval | Coverage | Mean width | Mean Winkler90 |
|---|---|---|---|
| Predictive-CDF shape | 347/384 (90.36%) | $7.86294 \times 10^{-5}$ | $1.02489 \times 10^{-4}$ |
| Symmetric log ratio | 351/384 (91.41%) | $1.21008 \times 10^{-4}$ | $1.29598 \times 10^{-4}$ |

At coverage close to the nominal 90%, the CDF interval is 35.02% narrower and improves mean Winkler score by 20.92%. The normalized paired Winkler difference is $-.831977$, interval $[-1.174406, -.543405]$, Holm $p = .00025$. Its lower/upper misses are 20/17, compared with 32/1 for the log-ratio interval: preserving skewness improves allocation of interval width. These gains concern uncertainty representation and calibration under the declared source mixture. The selected conditioner does not establish network-specific recovery beyond its activity-conditioned reference, or additional point-risk value over the source-supervised statistical readouts. Source coverage and market transfer are tested separately below.

Evidence record: calibrated-state study: joint primary contrasts, interval-score summary and frozen source-mixture design.

## 11 Connecting the network clock to real market observations

### 11.1 The earlier direct-map diagnostic

A decision unit, an execution and a calendar-time bar are different objects. A common-sign parent can generate multiple exchange executions; a five-minute bar can contain many internal decisions.

Financial transaction models treat arrival times and price changes jointly[17], providing a precedent for a separate observation clock. Here the original network continues to supply directional interaction, while the arrival/fragmentation layer introduced below maps them to market observations.

A descriptive transport check compares all 663 previously examined Bitcoin days, partitioned into 1,989 blocks of 96 five-minute bars, with 64 new 96-step source-network blocks. Table 7 summarizes scale-independent discrepancies under the tentative direct step/bar map.

**Table 7.** Uncalibrated direct-map observation comparison. Means concern the stated block summaries; three blocks share each market date and are not independent experimental replications.

| Block statistic | Source network | Bitcoin |
|---|---|---|
| Lag-one correlation of pressure $F/V$ | .709 | .142 |
| Coefficient of variation of volume | .150 | .928 |
| Coefficient of variation of count | .123 | .626 |

The direct map carries excessive directional persistence and too little activity variation. Raw exchange counts also lack a calibrated relation to 48 effective active units, and 31 real return bars exceed the unchanged source return cap. The model was therefore not evaluated as an independent new-market predictor. These diagnostics localize the observation problem rather than treating normalized features as a substitute for units, clocks and response calibration.

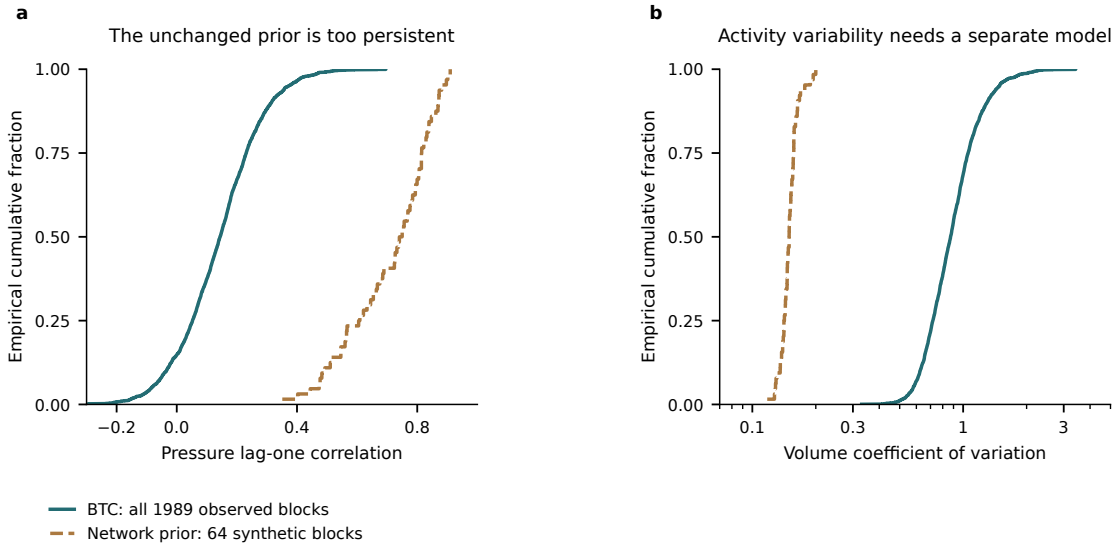



**Figure 14.** Source-network and Bitcoin observation checks under a tentative direct map. All 663 examined days and 64 new source-network blocks are used. Pressure persistence, activity variation, execution units and return support diagnose the observation layer before an independent market test is opened. Binance Vision-derived statistics retain source attribution.

## 11.2 What the calibrated source does and does not transfer to Bitcoin

The calibrated-state predictor was also evaluated at a fixed noon-UTC origin using 32 past five-minute $F, V, K, r$ bars to forecast the next 12 bars. The original internal market study fits empirical readouts on 181 H1 dates, selects penalties on 46 later dates and calibrates intervals on another 46. Its 390 evaluation dates had already been examined by the project. HAR-All and HAR augmented with the mechanism receive the same past observable fields, while the mechanism additionally uses its structural law, H1 response calibration and simulated future labels.

On those 390 dates, HAR-All has mean QLIKE .553812, HAR plus mechanism .546158 and HAR plus activity control .545175. The mechanism augmentation improves the HAR point score by 1.38%; the seven-day block paired difference is $-.007654$, interval $[-.013482, -.000867]$, Holm $p = .0459$. The result is sensitive to dependence scale (14-day $p = .0414$; 28-day $p = .0555$),

and the activity control has the better point score. Thus this internal result supplies neither coordination-specific predictive value nor independent confirmation.

After freezing code, predictor, deployment bank and calibration, 11 previously unused official BTCUSDT dates (26 October–5 November 2026) were acquired with verified archive checksums and five-minute time grids. All dates were retained, and predictions were persisted before joining their future outcomes. Mean QLIKE was .185205 for HAR-All, .186738 for HAR plus mechanism and .185148 for HAR plus activity control: the earlier augmentation advantage did not recur. This is a small frozen historical replay, not a prospective trading evaluation. All interval methods cover 11/11 outcomes, but no date exceeds the fixed high-risk threshold and the intervals are wide, so these dates do not establish tail-risk detection or reliable 90% coverage.

The stronger transport warning comes from direct use of the source-model CDF intervals on the 390 earlier Bitcoin dates: coverage is only 80.51%, with 41 dates beyond the fixed source-domain threshold. The 90.36% source coverage in Table 6 is consequently a useful model-inference result, not a market guarantee. Subsequent calendar-time experiments use these diagnostics to localize the remaining problem in the observable state and response map. The 11 replay dates later enter the expanded data used in the following state–environment studies; they are not an untouched confirmation set for those later results.

Evidence record: calibrated-state market readouts, source-to-market interval coverage and frozen unused-date replay.

### 11.3 Calendar-time observation and price-response extensions

The direct-map result above concerns the unchanged source law in Eq. (24). Subsequent work has implemented the separate observation layer it motivated. The retained 48-unit fast/slow network evolves through eight decision events per five-minute bar; an activity and execution layer produces calendar-time $F, V, K$, and the price layer separates realized impact from recovery of predictable impact. The M2 extension additionally supplies correlated residual decision innovations and feedback from actual calendar returns. These changes preserve the network mechanism while explicitly changing its observation and response law. Results under that extension are therefore reported separately from the earlier direct-map comparison.

The price-scale repair uses the population identity $\mathbb{E}[g^2] = 1$ for volume-scaled innovations, rather than an accidental finite-path average. Under the resulting nominal calibration, five-, fifteen- and sixty-minute return-second-moment ratios to Bitcoin are 1.063, 1.144 and 1.273; 30 of 36 fixed descriptive conditions pass. The nominal five-minute ratio had been about 1.570 in the preceding calibration. The revised model still misses pressure persistence, volume variation and the sixty-minute second moment. The impact coefficient's relative Monte Carlo standard error is 8.32%, above the declared 3% precision threshold, and the finite parameter ensemble does not improve its aggregate mismatch score. Thus a usable, frozen calendar-time generator has been constructed, while precise market calibration remains incomplete. Sections 12–13 use its nominal parameters to investigate conditional risk and the information problem rather than treating that generator as an identified real-market law.

The practical criterion remains lower total information cost at comparable reliability, or better accuracy with the same observable fields and history. If an extra liquidity observation is proposed, its identification role and its availability to every comparator must be explicit. The following studies first locate the state information that could justify that addition and then evaluate a macro-only approximation against observed Bitcoin risk.

Evidence record: earlier direct-map comparison V01; population price-scale repair and M1/M2 calendar-time continuation; source tables retained with the study records.

## 12 Coordination risk depends on the receiving environment

The weighted covariance mechanism in Eq. (14) determines how collective actions change flow variation. Price risk also depends on the environment through which that flow is transmitted. We now test this link on the retained weighted network in calendar time, rather than interpreting more coordination as a context-free volatility multiplier.

### 12.1 A common calendar clock and a controlled intervention

Let $\varrho$ denote correlation between future cross-agent residual decision innovations, $\kappa_r$ the coefficient of actual calendar-return feedback into decisions, and $L$ normalized latent shock capacity. These are different from the original ER degree $\rho$ and fast/slow layer coupling $\kappa$. A higher $L$ represents greater model capacity to absorb a shock; it is not an observed dollar order-book depth. The predictable-impact subtraction coefficient $\chi$ concerns the price response, separately from return-to-decision feedback.

The inherited calendar price equation can be written

$$r_j = \alpha + \beta I_j + \sigma\sqrt{g_j^2}\,\epsilon_j, \qquad g_j^2 = V_j/\overline{V}_{\text{fit}}, \tag{32}$$

where $I_j$ is realized impact minus predictable-impact recovery, formed before the current independent price innovation. The source records name this quantity $Y_j$; $I_j$ keeps it separate here from a risk outcome. Actual noisy returns still enter subsequent decisions. The risk target is $Z_{t,h} = \sum_{j=1}^{h} r_{t+j}^2$, with $h = 12$ five-minute bars, and $\mathcal{R}_h(S) = \mathbb{E}[Z_{t,h} \mid S]$ for complete state $S$.

A future-covariance intervention preserves every unit's initial complete decision signal by remapping its idiosyncratic component, then changes only the covariance of subsequent residual innovations. Paired arms share activity, execution and pricing innovations. Their $V, K$ paths and activity-scaled price-noise paths match exactly; all nuisance price parameters remain fixed. The comparison therefore asks how a change in future alignment propagates through the same initial market environment, rather than comparing different amounts of trading or refitting impact to each outcome.

### 12.2 Separating state heterogeneity from simulation noise

For each of 256 new independent M2 origins, three paired arms—full dynamics, future covariance off and return feedback off—generate 512 future branches. Origins are split 128/64/64 for fitting, development and evaluation. The first and second 256 branches provide independent labels and risk targets. We reduce current-bar label noise with

$$\mathbb{E}[r_j^2 \mid \mathcal{F}_j^{\text{pre-price}}] = (\alpha + \beta I_j)^2 + \sigma^2 g_j^2. \tag{33}$$

The tower property makes the sum of these conditional moments unbiased for expected future RV. Earlier price innovations still alter later states; this is termwise integration, not a claim that a global Rao–Blackwell variance-contraction theorem applies to the full feedback path.

The mean full-risk estimate is $3.6401\times10^{-5}$. Removing future covariance lowers it by $2.4830\times10^{-6}$, with paired-origin 95% interval $[2.2879, 2.6836] \times 10^{-6}$, or 6.82% of mean full risk. Removing return feedback lowers it by $1.5640 \times 10^{-6}$, interval $[1.2895, 1.8202] \times 10^{-6}$, or 4.30%. Integrated paired-label conditional variances are .299 and .283 times their direct squared-return counterparts; cross-half effect correlations rise to .911 and .974 (Fig. 15). Independent-replicate effect covariances are positive, so the measured state variation is not explained solely by Monte Carlo error.

These effects have a positive mean and real heterogeneity, but their marginal between-state variance is only about .28% and .50% of the full expected-risk variance. Those ratios are neither incremental $R^2$ nor causal shares of risk. More precise effect estimates still worsen evaluation QLIKE when jointly added to activity controls or simple-state controls, by 5.65% and 10.99%, respectively; both adjusted tests give Holm $p = 1$. The missing forecast value therefore cannot be attributed principally to too few future branches in the earlier study.

### 12.3 A conditional risk surface, rather than one coordination multiplier

A fresh response-surface study uses 32 origins and 128 paired future branches per cell. Its fixed grid combines $\varrho \in \{0, .35, .65\}$, $\kappa_r \in \{0, .15, .30\}$ and $L \in \{.5, 1, 1.5\}$ at $\chi = .5$, with two additional recovery settings at the nominal cell: 29 scenarios in total. Marginal cell intervals and paired intervention intervals answer different questions; Table 8 reports the latter.

The same future innovation alignment produces a larger price-risk increment when the simulated receiving environment has less capacity. The result gives an explicit mechanism for why a crowded

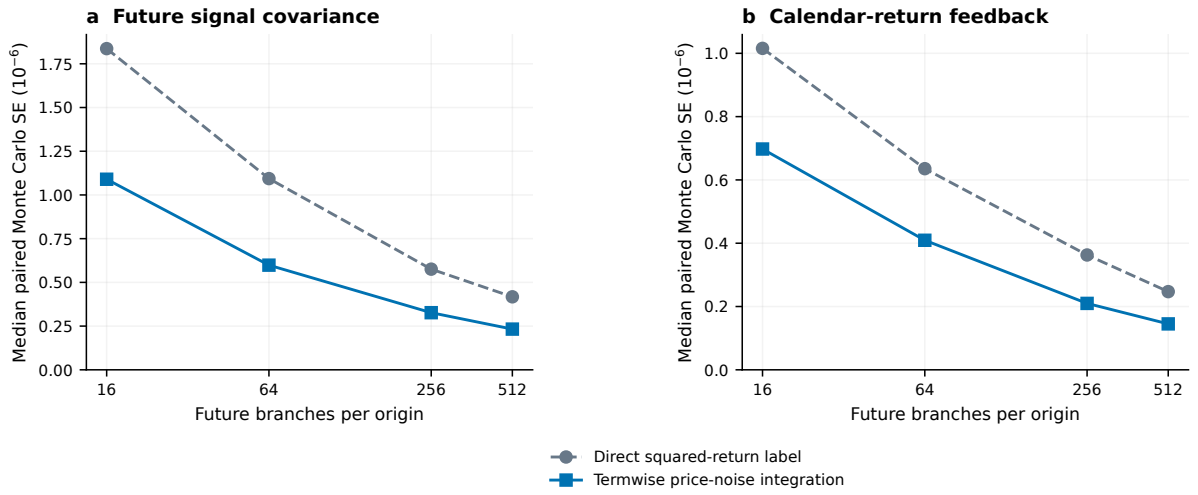



**Figure 15.** Independent-replicate precision diagnosis in the unchanged M2 dynamics. Each of 256 origins has 512 paired futures in each arm. Current-bar price-noise integration improves effect-label precision while preserving noisy future feedback. Average risk effects and cross-state heterogeneity are assessed separately from incremental forecast value. All quantities concern fixed nominal model parameters.

**Table 8.** Effect of increasing future innovation correlation from $\varrho = 0$ to .65 at $\kappa_r = .15$. Risk is the expected sum of twelve five-minute squared log returns. Intervals are paired-origin 95% basic-bootstrap intervals, with 4,999 replicates over 32 origins. Capacity is a latent model condition, not a measured market field.

| Capacity $L$ | Risk increment $\times 10^6$ | 95% interval $\times 10^6$ | Increment/full risk |
|---|---|---|---|
| .5 | 3.3063 | [2.7325, 3.8554] | 7.44% |
| 1.0 | 2.2303 | [1.6799, 2.7258] | 6.42% |
| 1.5 | 1.6389 | [1.1040, 2.0934] | 5.42% |

common response may matter especially during impaired market accommodation. It does not infer how many real institutions share a factor or the fraction of Bitcoin risk caused by them. Factor overlap would have to produce sufficiently weighted, directionally and temporally aligned trade innovations; the factor's name or the number of firms alone does not supply that measurement.

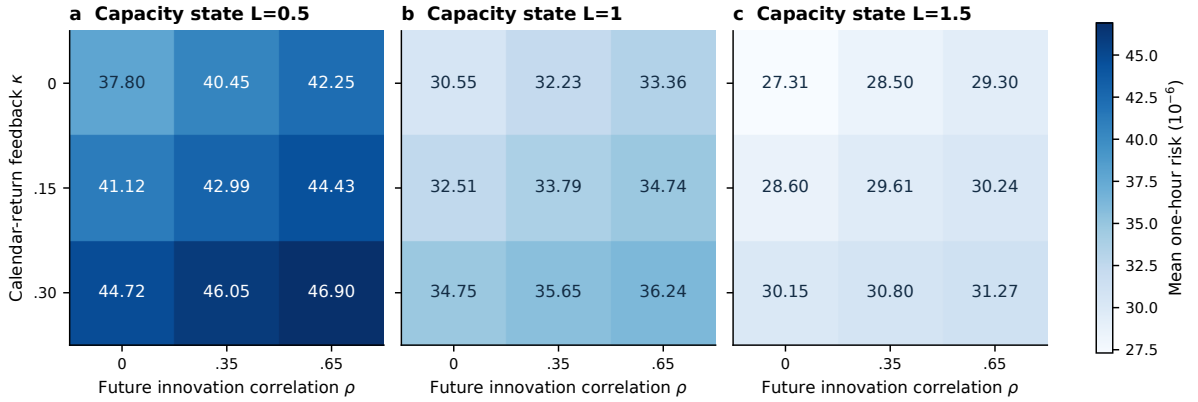



**Figure 16.** Expected-risk surfaces across fixed correlation, calendar-return feedback and capacity scenarios. The source data key `rho` corresponds to $\varrho$, the future innovation correlation, rather than the ER degree. Model states and interventions are distinct from observed Bitcoin quantities. All 27 main-grid cells and their source intervals are retained; two recovery cells complete the 29-scenario study.

The channels also interact. At $L = 1$, the covariance increment is $2.8116 \times 10^{-6}$ with $\kappa_r = 0$ and $1.4836 \times 10^{-6}$ with $\kappa_r = .30$. Their difference is $-1.3280 \times 10^{-6}$, paired interval $[-1.8572, -.7096] \times 10^{-6}$. Feedback raises total risk while reducing the marginal covariance effect on this grid. This is a conditional subadditive interaction, not a general saturation law or evidence that stronger feedback

is safe. Relative to $\chi = .5$, removing predictable-impact recovery raises risk by $2.8580 \times 10^{-6}$, interval $[1.4605, 4.0386] \times 10^{-6}$; increasing recovery to one does not establish an additional benefit. No optimal recovery rule follows from these three settings.

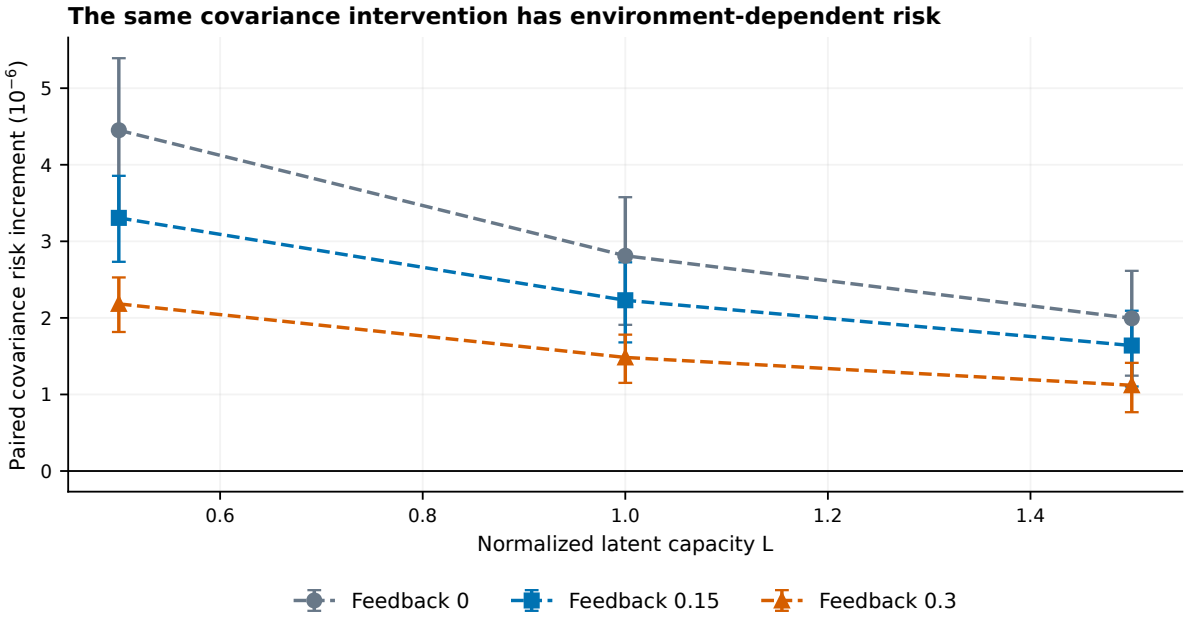



**Figure 17.** Paired covariance effects and environmental interactions. Error intervals resample whole simulation origins, preserving paired futures and shared paths. Lower shock capacity magnifies the tested covariance effect, while correlation and return feedback are not simply additive. Capacity, feedback and recovery are experimental model controls, not inferred policy thresholds.

For a fixed observed prefix, changing the future environment produces a median maximum/minimum expected-risk ratio of 1.694 across tested contexts. This is a useful stress-scenario range: it changes the future law and does not prove non-identifiability under one fixed stationary law. The explanatory object is consequently a state-and-environment risk map, not a unique coordination count recovered from one price–volume slope.

# 13 From mechanism effects to observable predictive states

## 13.1 Why a positive average effect is not a forecasting gain

Let $X$ be the allowed past macro history, $S$ the complete source state and $Z$ future realized risk. The paired contrast

$$\Delta(S) = \mathbb{E}[Z^{\text{common}} - Z^{\text{independent}} \mid S] \tag{34}$$

compares two intervention worlds. Prediction within one world instead asks which observations discriminate its higher- and lower-risk states. A nearly constant positive $\Delta$ changes the level absorbed by that world's baseline forecast. A variable $\Delta$ can be redundant with activity or remain unrecoverable from the permitted observations.

This distinction is exact for the conditional-mean target of QLIKE. For positive risk $Z$, forecast $a$ and $\ell(Z, a) = Z/a - \log(Z/a) - 1$, write $m_X = \mathbb{E}[Z \mid X]$ and $m_{X,W} = \mathbb{E}[Z \mid X, W]$ for additional state information $W$. Under finite expectations,

$$\mathbb{E}[\ell(Z, m_X) - \ell(Z, m_{X,W})] = \mathbb{E}[\log m_X - \log m_{X,W}] \geq 0. \tag{35}$$

Conditional averaging gives $\mathbb{E}[m_{X,W}/m_X] = 1$; conditional Jensen gives the inequality. The gain depends on within-world information about conditional risk, not the mean intervention contrast. Estimated learners need not attain this inequality, because finite-sample fitting and representation choices add error.

## 13.2 Which hidden state block adds information?

The independent-replicate design permits a direct information diagnosis. Each readout uses the same 128/64/64 origin split, a fixed Gamma/RBF learner menu and three penalty choices. Training labels use one set of 256 futures; evaluation targets use the independent second set. The target is average simulated future risk, which is less noisy than a single realized market outcome, so these losses are not directly comparable with the Bitcoin scores below.

**Table 9.** Effective-state information diagnosis at 64 evaluation origins. The hidden environment comprises day/intraday activity, size, liquidity and weight concentration. $R^2$ uses squared error relative to the fitting-period constant forecast, not the evaluation-period mean.

| Information supplied to readout | QLIKE | $R^2$ |
|---|---|---|
| Activity and past-volatility summaries | .028130 | .93171 |
| Complete 32-bar macro representation | .036879 | .87441 |
| Macro representation plus hidden environment | .008413 | .95498 |
| Above plus direction, memory and residual summaries | .008519 | .96631 |

The environment block improves QLIKE over the full macro readout by .028467, with signed difference interval $[-.038889, -.016976]$ and Holm $p = .0006$. Its joint gain cannot be attributed to activity alone: size, liquidity and weight concentration are included. Adding direction, memory and residual summaries establishes no further QLIKE improvement (Holm $p = 1$). The higher last-row $R^2$ measures a different error and does not reverse that result. Activity summaries nevertheless estimate covariance and feedback effect variation with evaluation $R^2$ about .749 and .863. Recovering an effect is possible without that effect adding information beyond a strong risk baseline.

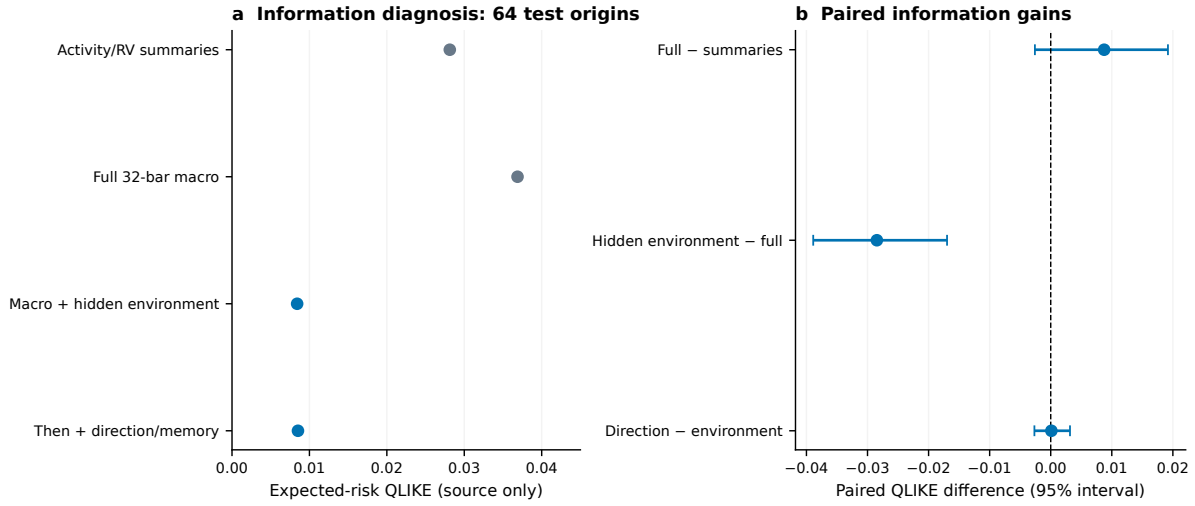



**Figure 18.** Risk information across matched source-origin and learner budgets. The environment block includes activity, size, liquidity and concentration; it is not an activity-only or observed-order-book intervention. Inferior performance of the full macro readout reflects finite-sample estimation, not a claim that supplying more raw information destroys information. Finite-neighborhood diagnostics are not treated as an identification impossibility theorem.

These results redirect the inverse task toward a joint effective state: pressure innovations, price response and environment-dependent residual risk. They do not support increasing simulation effort or adding another coordination scalar without a distinct information role.

### 13.3 An observable innovation–response closure

The network motivates a past-only macro approximation that separates persistence from new pressure:

$$p_t = F_t/V_t = \widehat{p}_{t|t-1} + u_t, \tag{36}$$

$$r_t \simeq \alpha + \lambda_o[p_t - \chi\widehat{p}_{t|t-1}] + e_t. \tag{37}$$

A pressure autoregression uses candidate lags 1, 4 or 12, previous actual return and clock terms; a finite fitting-period menu selects $\chi \in \{0, .5, 1\}$. A clock-aware log-volume/count model propagates activity. Local twenty-bar uncentered innovation second moments, with a zero conditional-mean approximation, supply a positive-semidefinite noise representation. First and second moments are propagated over the same twelve future bars through a linear state covariance recursion $\Sigma_{j+1} = A\Sigma_j A^{\mathsf{T}} + Q_j$, including direct first-bar innovation. This is an effective moment closure, not exact latent-state recovery in the nonlinear network.

Persistent order flow and compensating price/liquidity response motivate this separation[20, 21]. Those studies concern their own market observations and do not validate this particular five-minute Bitcoin approximation. Here the closure inherits the network's pressure, feedback and impact-recovery interpretation; its forecast value is judged by the same observed target as the statistical controls.

The fitting period contains 181 dates and 3,801 risk origins, plus 47,060 deduplicated past-bar rows for self-supervised one-bar fitting. The first 145 dates fit the inner model and the last 36 select its finite menu before refitting the full period. The first 46 dates of the following development quarter select a risk readout; the last 46 calibrate intervals. The selected pressure lag is 12 and selected recovery is $\chi = 1$. This menu choice does not identify a true Bitcoin recovery coefficient. All methods receive the same $F, V, K, r$ fields, thirty-two-bar history and fitting data; the mechanism adds derived representations, priors and computation, which remain part of its total cost.

### 13.4 Internal Bitcoin results and a representation control

At UTC 12:00 each day, the target is the next twelve five-minute squared returns. All 674 examined dates have appeared earlier in the project; the 401 scored dates are an internal reused evaluation, not new external confirmation. Seven fixed mechanism readouts and three Gamma penalties are developed without joining evaluation truth. Among RiskOnly, HAR+Risk and HAR+Components, development selects HAR+Components. Table 10 keeps that primary identity distinct from two controls added after the mathematical cancellation below was discovered.

**Table 10.** One-hour Bitcoin risk evaluation on 401 previously seen dates. Lower QLIKE is better. The first four rows belong to the fixed comparison; the final two are post-run descriptive representation controls and do not reset the primary test family.

| Method | QLIKE | Role |
|---|---|---|
| HAR-All | .532309 | Same-field reference |
| RawHistoryGamma | .539515 | Developed empirical pipeline |
| HAR+Risk | .524478 | Single-risk readout |
| HAR+Components | .501501 | Selected primary candidate |
| HAR+RV20 | .502745 | Post-run control |
| HAR+RawMoments20 | .499219 | Post-run control |

The selected candidate improves the QLIKE point value over HAR-All by 5.79%, but its paired loss difference is $-.030808$, with seven-day-block 95% interval $[-.078475, +.017982]$ and Holm $p = .33$. Fourteen- and twenty-eight-day blocks also fail to establish improvement. Six fixed primary questions use within-family Holm correction; paired circular bootstrap resamples whole dates in 4,999 replicates rather than treating overlapping windows as independent. The improvement over the developed raw-history pipeline is likewise uncertain (7.05% point value; Holm $p = .212$). The selected candidate and HAR-All both cover 94.51% at nominal 90%, with widths $1.1905 \times 10^{-4}$

and $1.1831 \times 10^{-4}$ and CRPS $1.2006 \times 10^{-5}$ and $1.2301 \times 10^{-5}$. These metrics do not establish overall distributional superiority.

One frozen ablation supports predictable-impact subtraction within the *single-risk* readout: HAR+Risk improves over HAR+NoRecovery by about .52%, with Holm $p = .0156, .0264, .024$ for 7-, 14- and 28-day blocks. This is not a causal decomposition of the selected HAR+Components result or an observation of actual market coordination. Feedback removal establishes no gain; the residual-channel comparison is not robust to the longer blocks. Full scores and contrasts remain in the source tables.

### 13.5 An exact cancellation that limits the forecast interpretation

An acceptance check identifies an informative limiting case. With $\chi = 1$, constant local innovation moments and *activity prediction switched off*, set $e_t = r_t - \alpha - \lambda_o u_t$, $v = \mathrm{mean}_{20}(u^2)$, $\eta_o = \mathrm{mean}_{20}(ue)/v$ and $z_t = e_t - \eta_o u_t$. Orthogonalization gives $\mathrm{mean}_{20}(uz) = 0$, so

$$(\lambda_o + \eta_o)^2 v + \mathrm{mean}_{20}(z^2) = \mathrm{mean}_{20}[(r - \alpha)^2]. \tag{38}$$

Complete recovery also removes predictable pressure from the future mean return in this closure. Its local-noise-only risk therefore becomes

$$\widehat{\mathcal{R}}_{12}^{\mathrm{local}} = 12 \left\{ \alpha^2 + \mathrm{mean}_{20}[(r - \alpha)^2] \right\}. \tag{39}$$

The identity is independently reconstructed on all 3,801 fitting and 674 query histories; maximum relative query discrepancy is $3.25 \times 10^{-15}$. It applies to this no-activity-forecast limit, not automatically to the full activity-scaled candidate.

This identity motivates the two explicitly post-run controls in Table 10. Adding ordinary RV20 nearly reproduces the main point improvement, and adding twenty-bar direct macro moments yields a lower point loss. They use fixed fitting/development procedures and do not become a new primary advantage test. The current improvement can consequently be explained by additional summaries or representations; it cannot be assigned uniquely to inferred coordination or network structure. The positive simulation mechanism, the useful conditional risk surface and this forecast boundary are compatible conclusions about different questions.

Source tables and methods: `studies/state_environment_v9/`.

## 14 Strategy scale, factor diversity and market accommodation

### 14.1 Shared strategy inputs extend the same weighted network

The weighted relation $\mathrm{Var}(F \mid Q) = Q^{\mathsf{T}} \Gamma Q$ makes common quantitative-strategy exposures a natural continuation of the herding model: correlated inputs change the action covariance, and effective trading weights determine how that dependence reaches aggregate flow. Strategy scale dependence and crowded exposure are established economic concerns[18]. Here their roles are separated inside the retained 48-unit fast–slow network, rather than introduced as a different forecasting mechanism. Graphs, update laws, memory, adaptation, activity and nominal calendar-time price parameters remain those of the preceding sections.

Let unit $i$ have positive normalized trading weight $w_i$, $\sum_i w_i = 1$, and belong to a factor-input group $g(i) \in \{1, \ldots, M\}$. Each unit has one positive group loading, a common loading and an idiosyncratic input:

$$u_i = \sigma_s \left\{ \sqrt{\rho_0} [\sqrt{c}\, Z_0 + \sqrt{1 - c}\, Z_{g(i)}] + \sqrt{1 - \rho_0}\, \epsilon_i \right\}, \tag{40}$$

$$\mathrm{Cov}(u_i, u_j) = \sigma_s^2 \rho_0 [c + (1 - c)\mathbf{1}\{g(i) = g(j)\}], \quad i \neq j. \tag{41}$$

All primitive processes are independent unit-variance Gaussian AR processes, with fixed $\phi = .6$, $\rho_0 = .65$ and $\sigma_s = .35$. The single-unit marginal signal variance and persistence stay fixed as $M$ and $c$ change. Here $c$ is the common fraction of factor innovations, not the proportion of quantitative

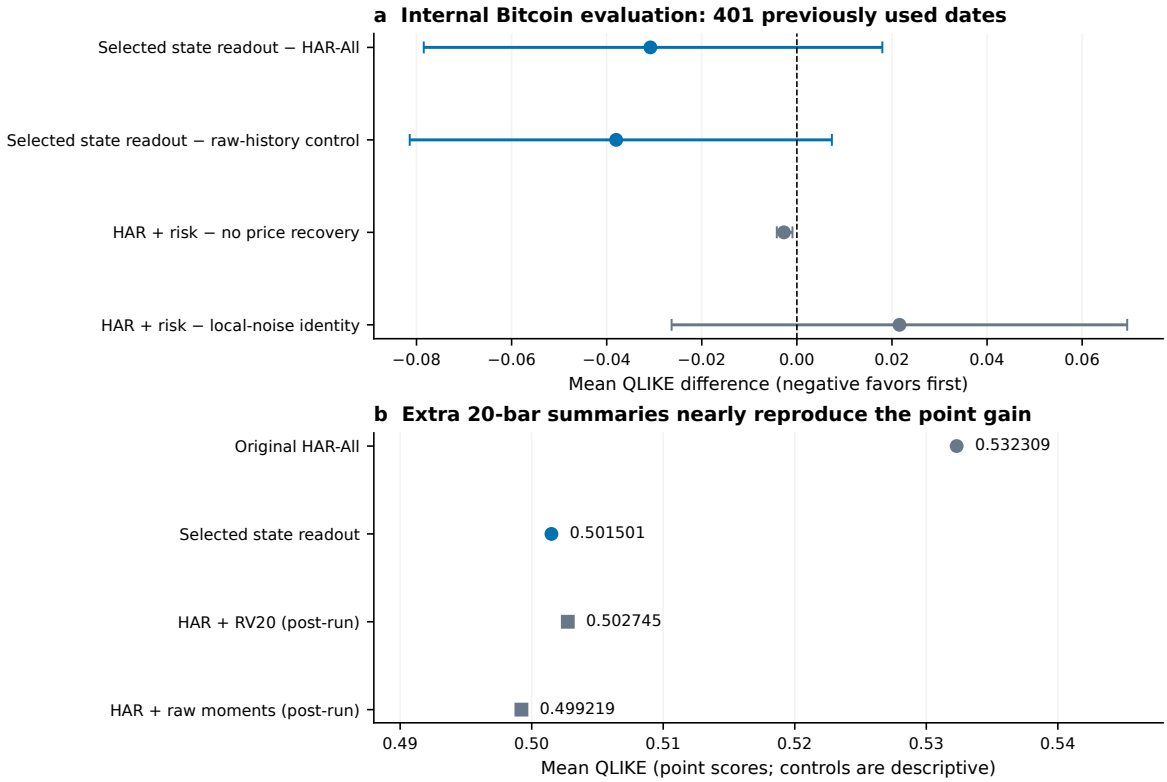



**Figure 19.** Observed internal Bitcoin forecast comparison and the closure's cancellation check. The selected candidate is distinguished from post-run RV20 and direct-moment controls. Evaluation dates are reused project observations; simulated environment cells are not superimposed as measured market states. The cancellation concerns $\chi = 1$ with constant local noise and activity prediction off.

traders. Twelve independent group inputs do not create twelve independent actors: their actions remain coupled through the original network and price feedback.

Define capital-weighted group exposure $P_g = \sum_{i:g(i)=g} w_i$, concentration $H_P = \sum_g P_g^2$ and effective group count $M_w = 1/H_P$. The next innovation in weighted input has conditional variance

$$\mathcal{V}_{\text{in}}^{\text{innov}} = (1-\phi^2)\sigma_s^2 \left[ (1-\rho_0) \sum_i w_i^2 + \rho_0 \{ c + (1-c) H_P \} \right]. \tag{42}$$

Risk-relevant diversity is therefore weighted exposure and correlation, not the number of labels. Equal numbers of nodes per group need not give equal $P_g$. At $c = 1$, every group is a clone of the same signal: relabeling $M = 1, 4, 12$ leaves complete simulated paths identical. For $c = 0$, nested group refinement reduces $H_P$ while preserving each node's input variance.

The intervention preserves the complete initial node signal by remapping its initial idiosyncratic component; common persistence then preserves the same initial conditional signal mean. It changes future innovation covariance from the same complete origin. These are conditional future scenarios, not separately equilibrated stationary strategy populations. The sparse positive-loading family is a transparent baseline; dense exposures and opposing factor loadings are outside this comparison.

## 14.2 A one-event mathematical bridge, followed by dynamic risk propagation

At a fixed complete pre-event state, let $H_i$ collect the known neighbor, slow-state, memory and calendar-feedback fields. For standardized Gaussian innovation $X_i$, the conditional mean of the updated fast sign is

$$t_i(X_i) = \tanh\left( 1.5 [H_i + \phi u_{i0} + \sigma_s \sqrt{1-\phi^2}\, X_i] \right). \tag{43}$$

With independent transition draws and independent Bernoulli($a$) activity, $\xi_i = B_i s_i$ obeys

$$\begin{aligned} \text{Var}(\xi_i) &= a - a^2 (\mathbb{E} t_i)^2, \\ \text{Cov}(\xi_i, \xi_j) &= a^2 \,\text{Cov}(t_i(X_i), t_j(X_j)). \end{aligned} \tag{44}$$

For standard bivariate Gaussian inputs with correlation $q$, the smooth bounded case of Price's identity[1] gives

$$\frac{d}{dq}\mathbb{E}[t_i(X_i)t_j(X_j)] = \mathbb{E}[t_i'(X_i)t_j'(X_j)] \geq 0. \tag{45}$$

The marginal means and diagonal action variances stay fixed, and every $w_i w_j$ is positive. Reducing pairwise Gaussian correlation therefore weakly reduces the next event's weighted flow variance. The implemented 4-to-12 grouping is a nested refinement, so no pairwise input correlation increases. Independent Gaussian quadrature checks both the covariance calculation and this ordering.

This is an application of a standard covariance identity to the present decision law. The fixed-field argument ends after one event: adaptation, changing signs, nonlinear impact recovery and actual-return feedback alter later marginals and fields. One-hour price-risk ordering must therefore be evaluated by continuing the full network, not inferred from Eq. (42) alone.

Effective scale $A$ multiplies the network's signed and unsigned trading quantities while leaving the independent background template unscaled. $A = .5, 1, 2$ is an order-weight intervention; .5 to 2 is fourfold network trading weight, not a doubling of firms, a measured AUM change or a quant-participation estimate. Capacity $L = .75, 1.3, 1.75$ sets both initial accommodation and its mean-reversion target. It is a normalized latent response condition, not observed order-book depth. Actual impact and predictable-impact recovery use the same intervention values.

For each future bar, actual noisy returns still enter later feedback. The primary one-hour risk is estimated by integrating only the current price shock:

$$\widehat{R}_{12} = \sum_{j=1}^{12} \left[(\alpha + \beta I_j)^2 + \sigma^2 g_j^2\right], \qquad g_j^2 = V_j / V_{\text{fit}}. \tag{46}$$

Averaging over complete futures estimates $\mathbb{E}\sum_j r_j^2$ by the tower property. The impact/drift term is not a coordination-risk share, and the statistic is not the variance of cumulative return. The 56 fixed scenario cells comprise 23 factor/network-ablation and 33 scale/capacity cases, each with 32 independent complete origins and 128 paired futures of 12 five-minute bars. The statistical unit is the complete origin. Each stage has eight declared contrasts, 4,999 paired-origin basic-bootstrap replicates and within-stage Holm correction. Intervals condition on the nominal generator; they do not absorb its market-calibration uncertainty.

### 14.3 Genuine diversification buffers risk; nominal diversity does not

At $A = 1, L = 1.3$, replacing a single common factor by four independent group factors reduces mean one-hour risk by 4.37%; twelve reduce it by 5.54%. For twelve groups, the absolute difference is $-2.0178 \times 10^{-6}$, 95% interval $[-2.5493, -1.4896] \times 10^{-6}$. Restoring a common share $c = .6$ raises risk by 3.94% at the same group count. The clone control is exact, not merely statistically indistinguishable: extra names for the same input cannot diversify the network.

**Table 11.** Locked one-hour scale/diversity/capacity contrasts. Percent change uses the reference mean in each row. Intervals have squared-log-return units, shown in $10^{-6}$. Every row has 32 independent origins; Holm $p = .0016$ within the eight-test stage.

| Treatment relative to reference | Change | Absolute difference [95% CI] |
|---|---|---|
| $A: .5 \to 2$, single common input, $L = 1.3$ | +119.63% | 32.570 [23.548,40.771] |
| $A: .5 \to 2$, 12 independent inputs, $L = 1.3$ | +102.04% | 27.195 [19.093,34.450] |
| 12 independent versus common, $A = 2$ | −9.95% | −5.950 [−7.149,−4.691] |
| 12 independent versus common, $A = .5$ | −2.11% | −.575 [−.730,−.402] |
| $c: 0 \to .6$, 12 groups, $A = 2$ | +7.65% | 4.119 [3.220,4.986] |
| $L: 1.75 \to .75$, 12 independent, $A = 2$ | +57.40% | 27.105 [19.603,34.014] |
| Full directional scale versus volume-only, $A = 2$ | +17.23% | 7.915 [4.889,10.597] |

At high scale, independent inputs provide a larger buffer: −9.95%, compared with −2.11% at low scale. The scale-by-diversity difference-in-differences is $-5.3750 \times 10^{-6}$, interval $[-6.4357, -4.2521] \times$

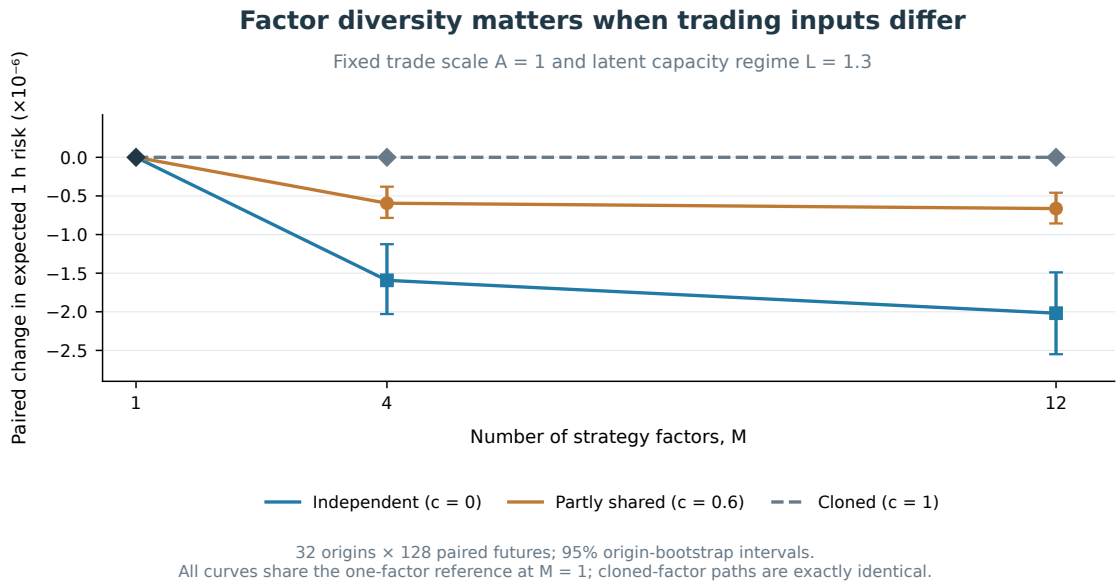



**Figure 20.** Conditional model scenarios for factor diversity on the retained weighted fast–slow network. Independent group inputs buffer one-hour price risk; perfectly common labels reproduce identical paths. Each case uses 32 complete origins and 128 paired futures. These are input-exposure interventions, not observations of quantitative institutions. Source: strategy-ecology factor study.

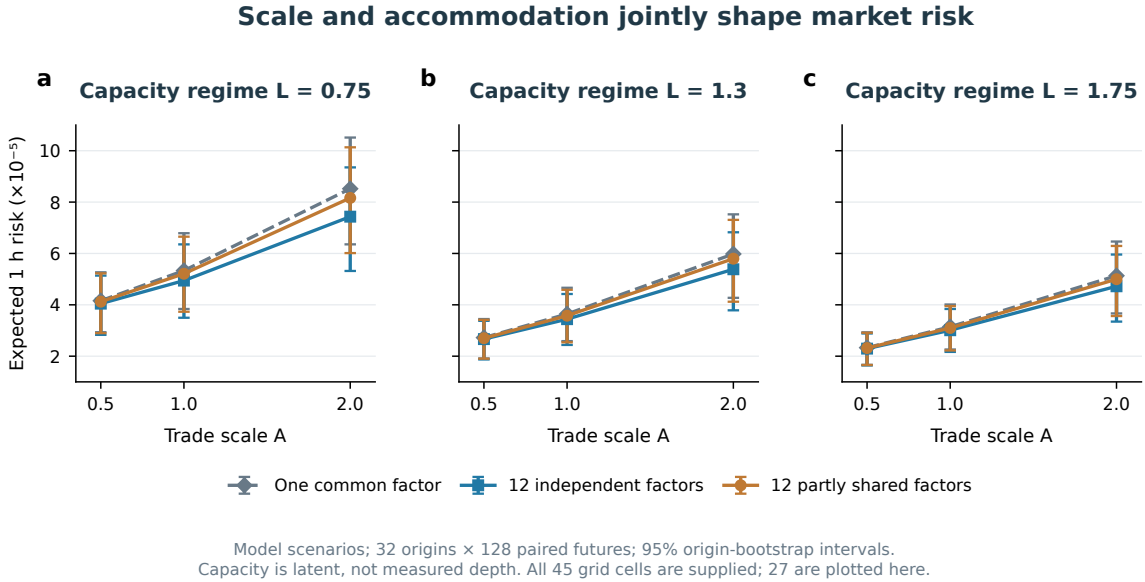



**Figure 21.** Model one-hour price risk across effective network trading scale, factor dependence and accommodation. Increasing independent factor diversity buffers scale expansion but does not fully offset the fourfold order-weight expansion in this fixed grid. Capacity is a latent model condition. Shared volume/count/noise controls distinguish directional pressure from extra activity. Source: strategy-ecology scale study.

$10^{-6}$; no arbitrary percentage denominator is assigned to this interaction. Even with twelve independent inputs, fourfold order-weight expansion increases risk by 102.04% under this grid. Diversification and scale thus act in opposite directions, and a factor count by itself cannot determine their net effect.

Part of scale's risk effect can come from more volume and activity-scaled price noise. The volume-only reference adds balanced executions at $A = 2$ while retaining the $A = 1$ net directional scale. Volume, count and price-noise paths are exactly matched to the full $A = 2$ arm. Full directional scaling still raises risk by 17.23%, interval [15.68%,19.27%] for the relative contrast. The directional-pressure channel consequently survives a control for the extra activity/noise channel.

This control is a synthetic balanced-execution diagnostic, not an observed counterparty balance sheet.

Accommodation can dominate this diversity buffer. At $A = 2, M = 12, c = 0$, lowering $L$ from 1.75 to .75 increases risk by 57.40%, relative interval [54.15%,60.13%]. Common-input, scale and capacity effects are conditional contrasts, not additive percentages of total market risk. The current evidence supports an environment-dependent mechanism rather than a universal quantitative-trading threshold.

## 14.4 Different interaction channels have different signs

The retained graph separates common inputs from neighbor influence, cross-layer coupling and feedback from actual returns. At $A = 1, M = 12, c = 0, L = 1.3$, turning off peer fields lowers one-hour risk by 4.73%; turning off actual-return feedback lowers it by 5.71%. Turning off fast–slow coupling instead raises risk by 7.14%, absolute difference $2.4559 \times 10^{-6}$, interval $[1.2490, 3.5612] \times 10^{-6}$. All three declared contrasts have Holm $p = .0016$. In this configuration, neighbor influence and return feedback amplify risk, whereas the net fast–slow interaction buffers it.

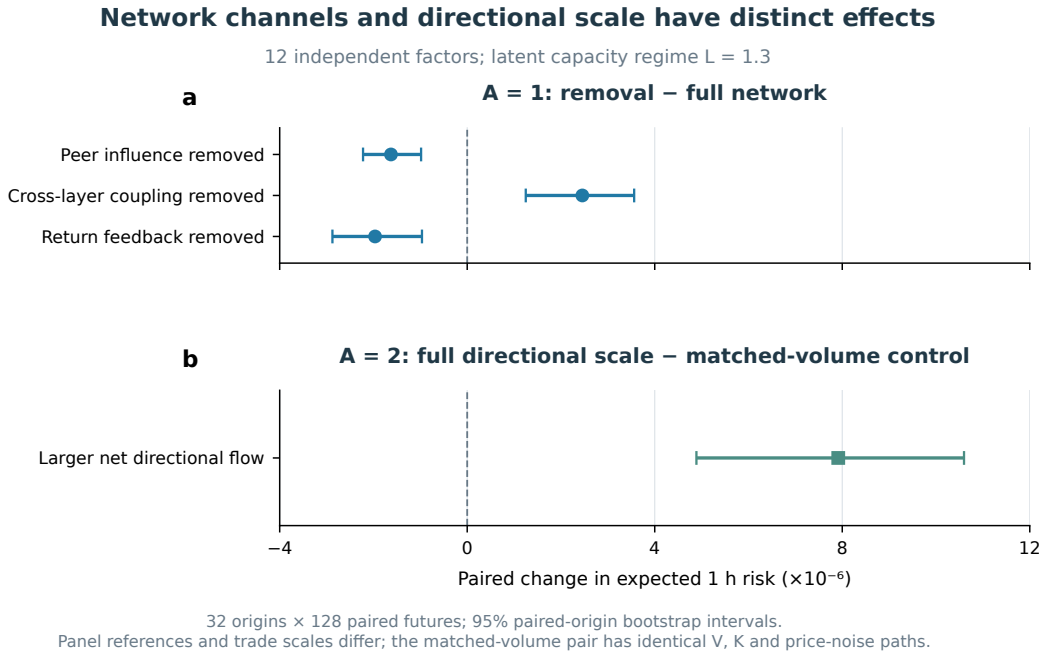



**Figure 22.** Separated model channels at the same factor exposure and nominal price law. Peer influence, fast–slow coupling and actual-return feedback have different signed effects. Turning off an interaction measures its net role in this configuration; it does not establish that every network connection amplifies risk. Source: strategy-ecology factor ablations.

The result strengthens the complex-systems interpretation: dependence changes risk through a dynamical response system, not through a rule that every connection is destabilizing. Shared input diversity remains useful even when actions are network-dependent. A positive mean mechanism effect also remains distinct from a precise state ranking: the low-scale diversity effect has half-sample origin correlation about .389. These scenario results supply explanatory risk responses; no new Bitcoin predictor or HAR advantage is claimed by this simulation stage.

Evidence record: strategy-ecology mathematical bridge, factor and scale primary contrasts, paired-path invariants and independent reconstruction.

# 15 Slow allocation: concentration and price risk are separate outcomes

## 15.1 Past performance changes future trading capacity

To test whether diversity can change endogenously, group shares now respond slowly to already completed performance. This extends the same factor exposures and network weights. At the preceding close, strategy $g$ holds a bounded shadow target $x_{g,t-1} \in [-1, 1]$ formed from its known fast directions. Its completed-bar payoff is

$$\pi_{g,t} = x_{g,t-1} r_t - 10^{-4} |x_{g,t-1} - x_{g,t-2}|. \tag{47}$$

The target and turnover are fixed before $r_t$ arrives. Every 12 bars, scores divide completed payoffs by the frozen H1 return standard deviation times $\sqrt{12}$ and are clipped to $[-2, 2]$. Allocation for the next bar is

$$P_g^{\text{new}} = .98 \, \frac{P_g^{\text{old}} \exp(\eta s_g)}{\sum_k P_k^{\text{old}} \exp(\eta s_k)} + .02 P_g^{\text{initial}}, \quad \eta = .10. \tag{48}$$

The 2% reinjection prevents absorbing zero shares. Initial within-group relative node weights are retained, and both capital-weighted neighbor matrices are refreshed on their unchanged graph supports. Shares remain positive and sum to one; total scale $A = 1$ is fixed. A slower-learning sensitivity uses $\eta = .025$.

Four- and twelve-group independent-input cases compare fixed, performance, score-identity-permuted and slower allocation, at $c = 0, L = 1.3$. Each of the eight cells has 32 complete origins, eight paired futures and 192 five-minute bars (16 hours). Permutation reassigns that arm's own completed scores to strategy identities, and can have fixed points. It preserves score scale and timing, rather than creating an independent or information-free return process. Its role is to test the score-to-identity link.

This rule reallocates next-period trading capacity and influence. Shadow payoffs are not a self-financing wealth account, and share changes do not create immediate rebalancing orders, margin calls or liquidation. These choices define the economic mechanism being tested; real capital migration can involve those additional channels.

## 15.2 Concentration increases, but a sixteen-hour risk increment is not established

**Table 12.** Performance allocation minus fixed allocation under the specified rule. HHI and RV are separate declared outcomes, with paired-origin inference. Risk differences and their intervals are in $10^{-6}$ squared-log-return units.

| Outcome | Four groups | Twelve groups |
|---|---|---|
| Terminal group HHI, relative change | +1.778% | +4.207% |
| HHI absolute difference | .004566 | .003905 |
| HHI 95% interval | [.003215,.005832] | [.003241,.004589] |
| HHI Holm $p$ | .0016 | .0016 |
| Mean pre-bar HHI, relative change (descriptive) | +.985% | +2.272% |
| Sixteen-hour expected RV, relative change | +.0382% | +.0228% |
| RV absolute difference | .2523 | .1488 |
| RV 95% interval | [−1.3785,1.7391] | [−1.4215,2.0302] |
| RV Holm $p$ | 1 | 1 |

Performance-based allocation increases terminal concentration relative to both fixed and score-permuted references. Against permutation, HHI Holm $p$ is .0096 for four groups and .038 for twelve. The corresponding risk differences are +.174% and +.158%, but their intervals cross zero (Holm $p = .2176, .8598$). Thus the allocation rule has a supported concentration effect without an established price-risk increment over this sixteen-hour window.

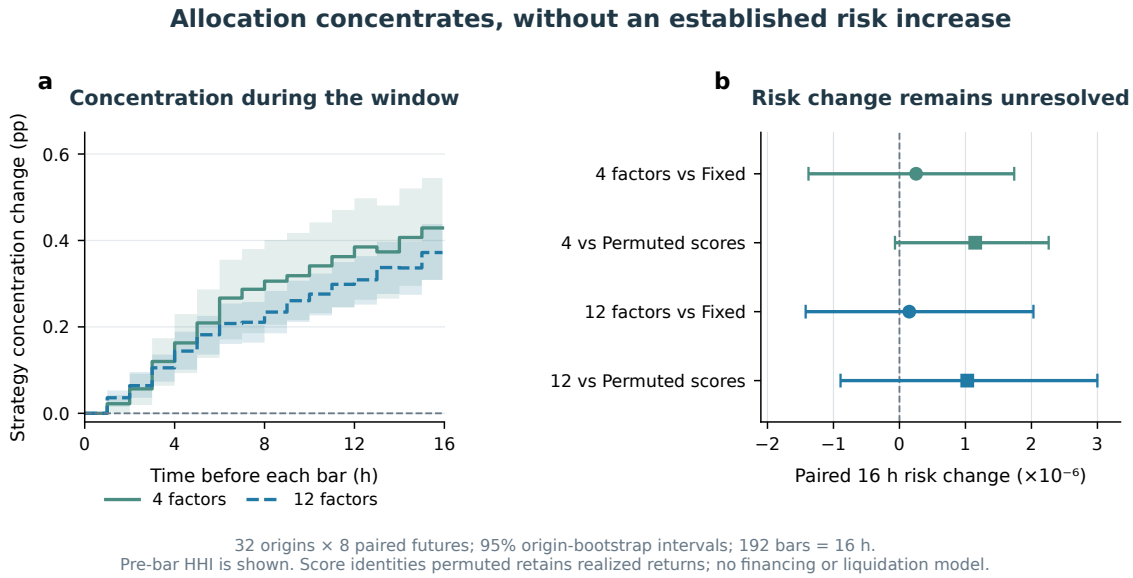



**Figure 23.** Slow allocation in the model: strategy concentration increases, while the sixteen-hour price-risk increment is unresolved. HHI and RV use different units and are shown separately. Terminal HHI follows the last update after bar 192 and describes bar 193 onward; pre-bar paths show concentration during the risk window. The score-permuted control uses its own arm's completed scores. Source: strategy-ecology allocation study.

Chronology matters for the mechanism interpretation. Terminal HHI is measured after the bar-192 update, and governs the next bar outside the accumulated risk window. It cannot be presented as a pre-existing mediator of already completed risk. Concentration also rises within the window: the mean pre-bar HHI moves from .256721 to .259250 for four groups and from .092816 to .094925 for twelve. Complete paths and slower-learning diagnostics appear in Fig. 36.

Equation (42) predicts more next-step input innovation variance from greater concentration when $c = 0$ and other terms are fixed. Terminal input innovation variance indeed rises by about 1.80% and 4.18% descriptively. That is an input-layer result, not the measured price-risk outcome. Nonlinear decisions, impact recovery, background trading and activity-scaled noise all intervene before multiperiod risk. This experiment does not uniquely identify which transmission channel accounts for the price-risk null. It shows why concentration is an informative structural diagnostic that still requires a horizon-matched risk test. The finite-window null establishes neither long-run safety nor risk equivalence.

Evidence record: strategy-ecology allocation protocol, primary contrasts, within-window HHI paths and independent replay.

# 16 Observed factor returns test the exposure-to-risk bridge

## 16.1 Research-factor observations and their scale

The model makes exposure covariance central, but its factor groups have positive sparse loadings and controlled Gaussian inputs. Observed research factors provide a direct test of which parts of that logic survive signed market exposure. We use the official Kenneth R. French Data Library's daily five-factor and momentum series[2]: market excess return, size, value, profitability, investment and momentum. These are returns on constructed research portfolios. They are observed data; the allocations and risks below are computed from them. They do not identify quantitative firms, their orders or real fund flows.

The common panel contains 9,254 trading dates from 2 January 1990 through 30 September 2026. Month-end origins require 252 previous daily observations and mechanically complete next calendar months. The resulting 429 months split into 228 fitting months (1991–2009), 120 development months (2010–2019) and 81 evaluation months (January 2020–September 2026). The protocol and code were frozen before this data acquisition; each allocation and prediction uses only its origin's

past. The data are the currently downloadable historical vintage, not a point-in-time archive. This is a frozen historical evaluation, not a prospective institutional test.

At each origin, past-252-day standard deviations $s_g$ scale the six sleeves. The fixed rule uses risk budgets $b_g = 1/6$ and positive coefficients

$$a_g = \frac{b_g/s_g}{\sum_k b_k/s_k}, \qquad Y_{m+1} = \frac{1}{n_{m+1}} \sum_{d \in m+1} (a^\mathsf{T} f_d)^2. \tag{49}$$

Coefficients are held fixed over the next month. $Y$ is a mean daily squared return of a factor-sleeve combination, not intraday RV or a funded institution's NAV. The signed underlying long/short factor exposures allow hedging even though sleeve coefficients are positive.

The full-covariance predictor uses the last 63 daily factor returns, their sample-centered covariance with divisor 63 $\widehat{\Sigma}$ and mean $\widehat{\mu}$:

$$\begin{aligned} \widehat{R}_{\text{full}} &= (a^\mathsf{T}\widehat{\mu})^2 + a^\mathsf{T}\widehat{\Sigma}a, \\ \widehat{R}_{\text{diag}} &= (a^\mathsf{T}\widehat{\mu})^2 + \sum_g a_g^2 \widehat{\Sigma}_{gg}. \end{aligned} \tag{50}$$

Both preserve the same means and marginal variances. Their difference is the signed cross-covariance term $2\sum_{g<h} a_g a_h \widehat{\Sigma}_{gh}$. The effective number of correlation modes is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$, for eigenvalues of the past-63-day correlation matrix. This observable quadratic-form test uses the same weighted-covariance principle as the network; it does not fit or validate the network's latent capacity parameter.

### 16.2 Six factor names do not determine six independent risk sources

During the 81 evaluation months, the effective correlation-mode count has median 3.228, range 1.900–4.448, compared with six labels. The signed covariance contribution is negative in 36 months and positive in 45. Full-covariance divided by diagonal estimated risk ranges from .355 to 2.301, with median 1.021. Dependence can therefore hedge or amplify the exposure-weighted risk. These are past-window diagnostics, not causal estimates of coordination or its share of future risk.

This observation refines the model's positive-loading benchmark. Equation (42) correctly isolates common positive-input dependence under its stated construction. Real factor combinations require the full signed covariance matrix; a scalar common-factor share or nominal factor count cannot replace it. Extending the network to signed and dense factor exposures is consequently an empirical requirement, not a change to the macro-to-structure-to-risk main line.

### 16.3 Forecasting with shared observable histories

The statistical controls receive the same six-series 252-day past and fitting/development periods. HAR-Portfolio uses 22-, 66- and 252-day second moments of the current coefficient combination. HAR-All adds individual-factor second moments, recent means and all 15 pair correlations. Both are positive Gamma log-mean ridge models fitted on 1991–2009; a fixed three-penalty grid is selected only on 2010–2019, with no evaluation refit. Equal observable histories do not make the representations or parameter counts identical.

**Table 13.** Frozen historical factor-risk prediction, 81 evaluation months. All methods predict Eq. (49). Lower QLIKE is better. HAR-Portfolio is retained even though it is outside the two declared predictor contrasts.

| Method | Mean QLIKE |
|---|---|
| Full signed covariance | .242664 |
| Diagonal covariance | .349737 |
| HAR-Portfolio | .193406 |
| HAR-All | .488429 |

Full covariance improves the diagonal point score by 30.62%, but its paired QLIKE difference $-.107072$ has three-month-block 95% interval $[-.201104, +.055958]$ and Holm $p = .2562$. Against

HAR-All the point improvement is 50.32%, difference −.245764, interval [−.386143, −.022985], but six-test Holm $p = .0810$. The prespecified six-month block sensitivities give $p = .3088$ and .0850, respectively. The simpler HAR-Portfolio has the lowest point loss. The study therefore supports a transparent signed-covariance diagnostic, while robust superiority over the supplied forecasting references is not established.

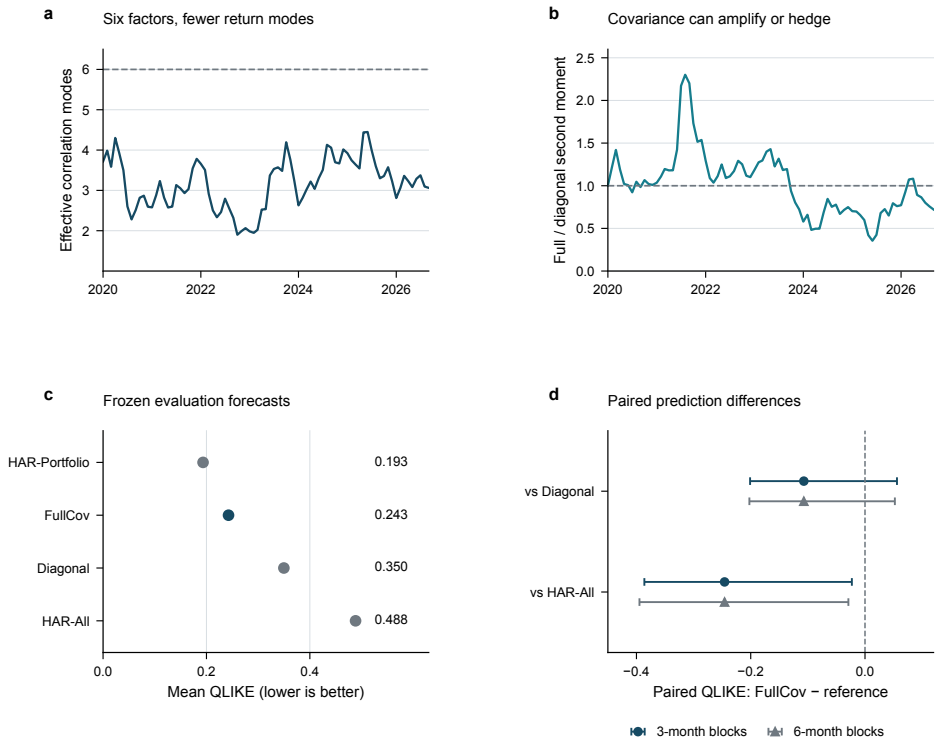



**Figure 24.** Observed research-factor returns and computed risk diagnostics. The six nominal labels correspond to fewer effective correlation modes; signed covariance hedges or amplifies exposure risk. Predictor contrasts retain paired block uncertainty and the stronger simple HAR-Portfolio point score. Estimates use only past data at each monthly origin, on the current historical data vintage. Source: official French factor panel and frozen factor-risk validation.

## 16.4 Performance allocation does not uniquely explain concentration

A monthly analogue of Eq. (48) updates risk budgets from the just-completed calendar month's factor returns, standardized by current past volatility and the square root of completed trading days, then clipped to $[-2, 2]$. Learning rate .10 and 2% equal-budget reinjection are fixed. The updated budgets are inverse-volatility transformed by Eq. (49); those actual coefficients, not raw budgets, define held HHI and next-month returns. The score-permuted control shuffles completed factor-score identities using a deterministic date-seeded permutation. Primary risk is gross; coefficient turnover and a descriptive one-basis-point charge are recorded separately, without fabricating unknown within-factor execution costs.

Mean held HHI is .189668 for fixed inverse-volatility allocation, .207024 for performance allocation and .229866 for permutation. Performance minus fixed is +.017357 (+9.15%), with three-month-block interval [.007006,.027132] and Holm $p = .0036$. Under the six-month sensitivity, Holm $p = .0540$. Performance allocation is thus more concentrated than fixed allocation in the primary analysis, but not more concentrated than its score-permuted control: the latter has the larger point HHI, and their adjusted comparison is unresolved ($p = .1624$). This blocks an attribution of all observed concentration to correctly directed performance feedback.

The performance rule's next-month realized-risk point estimate exceeds fixed allocation by 34.82%, but the absolute difference $4.1489 \times 10^{-6}$ has interval $[-2.9295, 8.2522] \times 10^{-6}$, Holm $p = .2562$. The difference from permutation is likewise unresolved ($p = .4844$). Six primary comparisons use 4,999 paired circular-block resamples and Holm correction together; months, not daily returns, are

the analysis units. Nonoverlapping target months avoid mechanically overlapping forecast outcomes, while block sensitivity addresses remaining serial dependence.

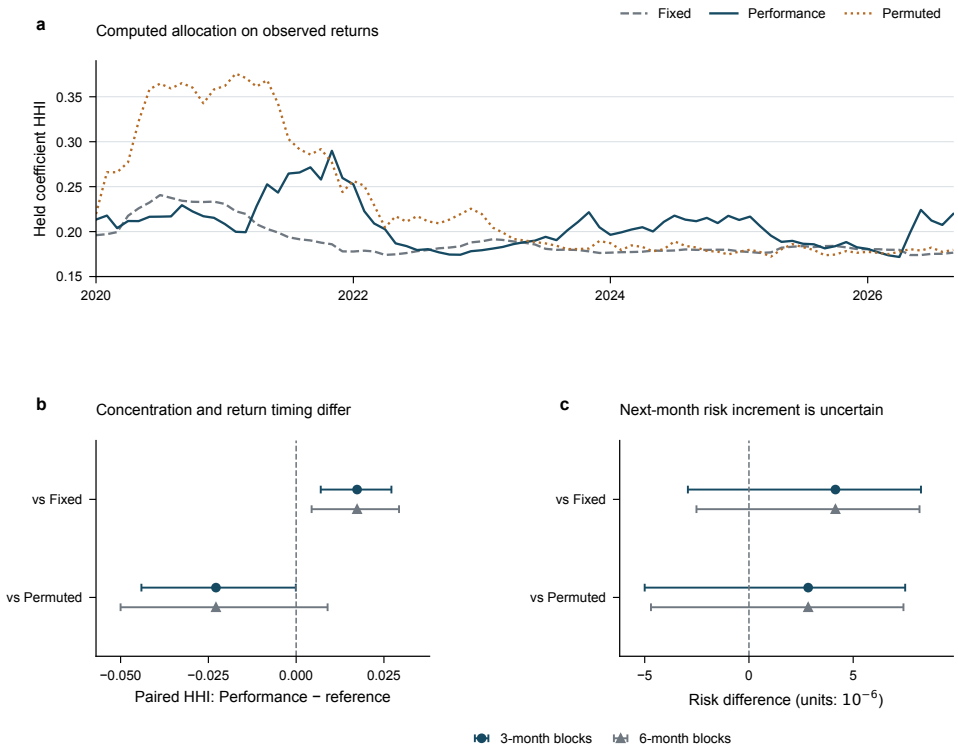



**Figure 25.** Computed held coefficients on observed research-factor returns. Concentration and next-month risk are separate outcomes, with three- and six-month block sensitivity. Performance allocation is more concentrated than fixed inverse-volatility allocation in the primary test, but the score-permuted control has higher point concentration; risk increments remain unresolved. These allocations are calculated scenarios, not observed fund flows. Source: frozen factor-return allocation evaluation.

This observed-return experiment adds a specific contribution to the original mechanism: effective exposure depends on signed covariance rather than factor labels; allocation can change held concentration without a robust risk or prediction increment. It does not establish that actual quantitative-trading growth causes market volatility. The market data have no endogenous feedback from these computed allocations, and contain no institutional impact or fund-flow observations. That feedback remains the role of the controlled network scenarios, while the historical factor panel constrains which exposures and controls a future market-calibrated version must represent.

Evidence record: official French factor archives, frozen acquisition/prediction records, complete 81-month contrasts and dependence sensitivities.

# 17 Historical next-day statistical forecasting

This benchmark concerns historical next-day Bitcoin volatility, distinct from both the one-hour network/Bitcoin risk and monthly factor-return targets above. Scores from these horizons and data designs are not pooled.

## 17.1 How physics and forecasting are connected

The physics motivates observable activity, joint price–volume statistics, and extreme-concentration measures. Market tests assess the fixed interface and reveal temporal dependence. The forecasting application directly uses these observables and multiscale lags, without true network or cluster labels. The heterogeneous autoregressive (HAR) representation follows the multiscale approach of Corsi[10]. It is an empirical forecasting specification using mechanism-motivated observables and temporal lags, with forecast accuracy evaluated against named statistical references.

After the final five-minute bar of day $d$ closes, the model forecasts the next UTC day:

$$RV_d = \sum_{j=1}^{288} r_{d,j}^2, \quad z_{d+1} = \log RV_{d+1}, \tag{51}$$

$$\widehat{z}_{d+1} = \bar{z}_{\text{fit}} + \widehat{w}^\top \widetilde{x}_d,$$
$$\widehat{w} = \arg\min_w \sum_{d \in \text{fit}} (z_{d+1} - \bar{z}_{\text{fit}} - w^\top \widetilde{x}_d)^2 + \alpha_R \|w\|^2. \tag{52}$$

Standardization uses the fitting period only. The ridge penalty $\alpha_R$ differs from a tail exponent. The displayed $100\sqrt{RV}$ measures intraday volatility intensity, without specifying the direction of the close-to-close return.

**Table 14.** The full model's ten inputs, all available at the forecast origin.

| Group | Definition | Count |
|---|---|---|
| HAR | $\log RV_d$; log mean RV over the past 7 days; log mean RV over the past 28 days | 3 |
| Activity | Log total daily BTC volume; log ratio of daily volume to its past-7-day mean | 2 |
| Coupling | Bar-level Spearman$(\lvert r\rvert, v)$; $\overline{r_c^2 v_c}/(\overline{r_c^2}\,\bar{v})$; absolute summed signed-taker volume divided by daily volume | 3 |
| Tail | $\log[288 \sum r_j^4/(\sum r_j^2)^2]$; $\max r_j^2 / \sum r_j^2$ | 2 |

$r_c, v_c$ are demeaned within a day. Signed activity derives from venue taker fields, not full order-book OFI or market-wide buy–sell imbalance. Tail inputs measure finite-sample shape and concentration, not power-law exponents. The 1/7/28-day horizons adapt the setup to a 24/7 market.

The original protocol used 337 eligible training targets in 2025 and 90 validation targets in 2026 Q1. After selection, one refit used all 427 targets. Parameters remained fixed across 183 evaluation dates from April 1 to September 30, 2026. Observed test-day data can become later lagged inputs, but do not enter refitting or renewed selection. Figure 26 retains every test date.

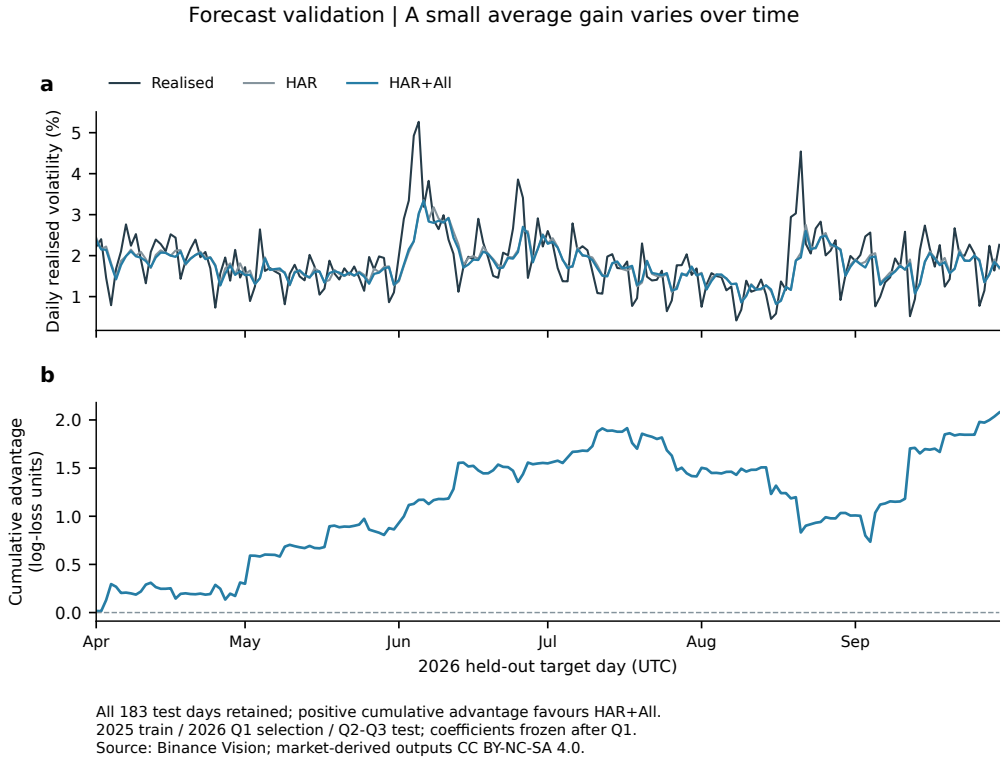



**Figure 26.** Forecasts and realizations on all 183 historical evaluation dates. Volatility is $100\sqrt{RV}$. The lower panel accumulates forecast-loss advantage over HAR. Binance Vision, CC BY-NC-SA 4.0. [P01]

## 17.2 Overall skill and incremental value are separate

For $D$ evaluation days, the primary loss and relative skill are

$$L_{\log} = D^{-1} \sum_{d} (\log RV_d - \widehat{z}_d)^2,$$
$$\text{Skill}_B = 100(1 - L_{\text{model}}/L_B). \tag{53}$$

Table 15 and Fig. 27 place simple and strong references together. The full model has positive historical skill over a constant, persistence, and EWMA. HAR alone already improves over persistence by 29.61%. The additional 2.263% point gain over HAR has an interval crossing zero, with Holm-adjusted $p = 0.3948$ in the original primary comparison. This contrast measures the combined addition of return-shape and trading-information features.

**Table 15.** Original next-day log-RV evaluation: 183 BTCUSDT dates.

| Reference/model | log-MSE | All loss reduction (%) | Approx. 95% CI (%) |
|---|---|---|---|
| Fit-period mean | 0.793999 | 38.26 | [24.65, 48.97] |
| Persistence | 0.712561 | 31.20 | [21.76, 38.62] |
| EWMA | 0.606913 | 19.23 | [13.22, 24.99] |
| HAR reference | 0.501565 | 2.26 | [-0.11, 4.56] |
| HAR+All | 0.490213 | — | — |

All denotes the original HAR+All. Intervals use 4,999 paired circular seven-day blocks, unadjusted, in a post-analysis overall-skill reassessment. Source: predictive_utility_v1.

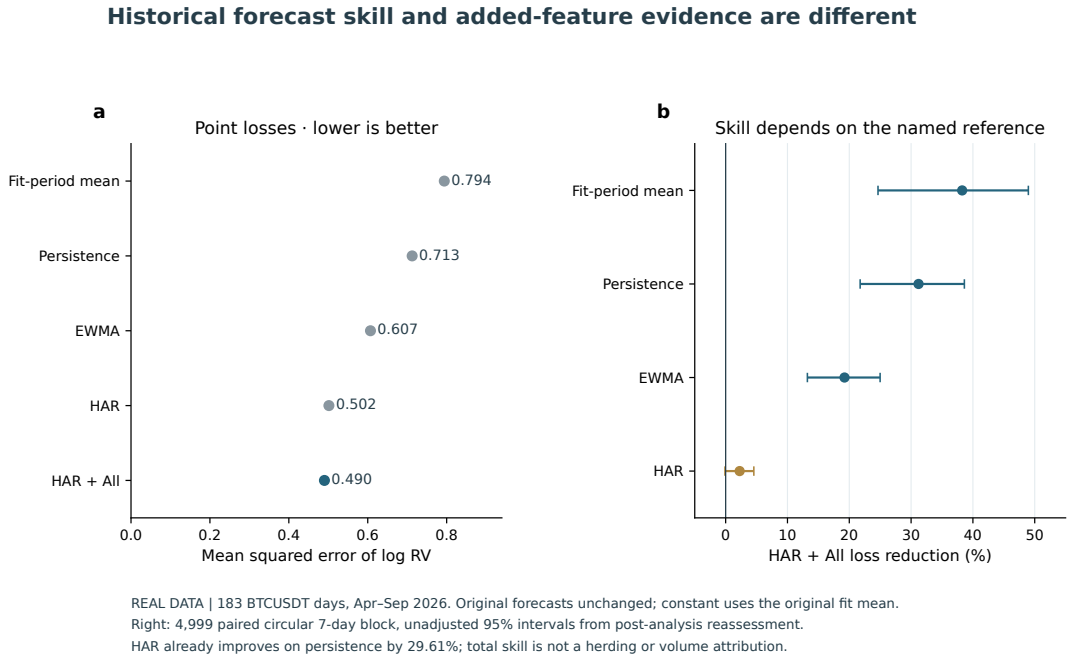



**Figure 27.** Historical log-MSE and full-model loss reductions against named references. Intervals use 4,999 paired circular seven-day-block resamples. The constant-reference comparison belongs to the post-analysis reassessment; the original HAR comparison is reported in the text. [J02]

The output is a next-day volatility estimate paired with a date-by-date error record. Subsequent asset and time tests in Appendix D examine the transfer of the added features. Prospective calibration and decision-specific losses would be needed to assess operational risk use.

## 17.3 Mechanism-imposed risk under a matched information budget

The statistical feature benchmark above asks whether selected observables improve a HAR forecast. A separate mechanism test asks whether a pressure-state restriction adds information while

preserving the same raw inputs and total-history budget. Both questions are retained, with their own protocols and scores.

The rolling mechanism study uses the same past price, volume, count and signed-flow fields as its macro-HAR reference. A reduced pressure model forecasts future flow moments and passes them through the response/risk map. At a 240-day total-history budget, its QLIKE is .260805; a same-map pressure AR control gives .260792, return-HAR .228191, and macro-HAR about .2236. The near identity with the pressure control means the tested network-motivated reduction supplies no demonstrated increment at this budget. The historical dates have been examined in prior project rounds and are development evidence, not a newly untouched test.

Further joint-flow-moment correction is evaluated at 30, 60, 120 and 240 days rather than selecting one favorable budget. At 240 days its QLIKE is .2749, versus .2236 for macro-HAR. Separately estimating residual cross moments produced incompatible covariance fits; the common-path construction in Section 10 addresses that calculation issue without retroactively changing these market scores.

A signed-flow/response bridge also generates sequential $\Delta = -9.562$, compared with $-8.181$ observed, while its selected additional forecast channel is Off. Its matched-HAR QLIKE .183724 and selected shared hybrid .186232 belong to that different fixed protocol. Thus explaining a negative compatibility statistic and improving a next-day forecast are separately evaluated achievements. Scores from distinct protocols are not interpreted as a chronological improvement series.

Evidence record: matched-budget collective-dynamics scores, joint-flow W3 comparisons and the chronological signed-flow bridge benchmark.

# 18 Portfolio and strategy risk diagnostics

## 18.1 How much sample risk can common movement conceal?

The US-equity study contains nine sector ETFs, SPY and QQQ references, and 46,530 daily rows through October 23, 2026. Financial association networks have established precedents[11]; our sector graph represents observed co-movement, not trader interactions. The original design uses 2010–2017 for training, 2018–2019 for selection, 2020–2025 for historical evaluation, and 2026 for confirmation.

For within-window return covariance $\widehat{\Sigma}$ and weights $w$,

$$\widehat{v} = w^\top \widehat{\Sigma} w = \underbrace{\sum_i w_i^2 \widehat{\sigma}_i^2}_{v_{\text{self}}} + \underbrace{2 \sum_{i<j} w_i w_j \widehat{\sigma}_{ij}}_{v_{\text{collective}}} . \tag{54}$$

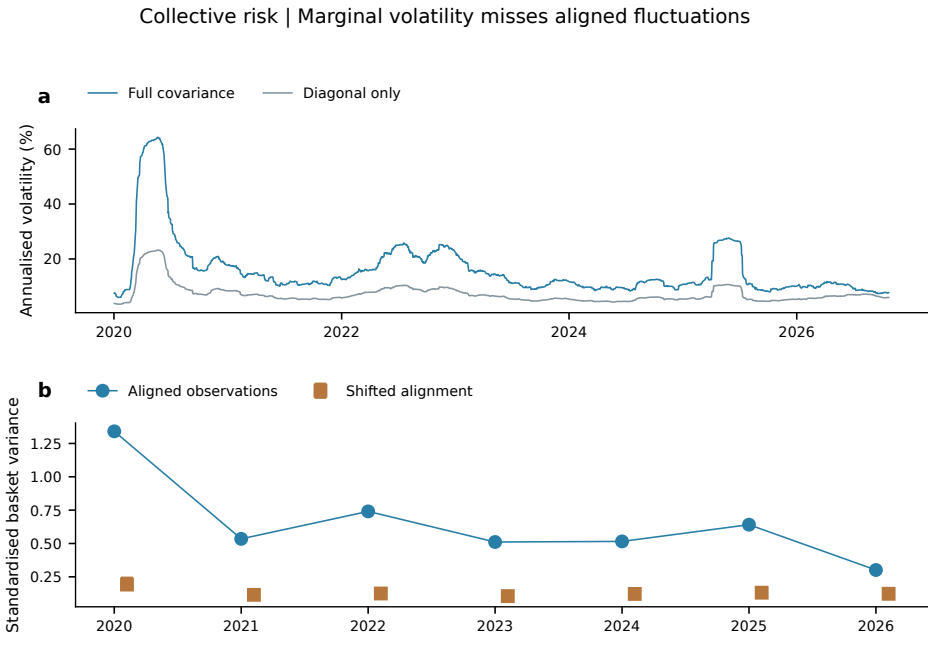
Annualized volatility is $100\sqrt{252\widehat{v}}$. For nine equally weighted sectors in the 63-day window ending March 31, 2020, the full estimate is 57.73%. Retaining only diagonal terms gives 20.62%, a ratio of 2.80. Collective terms account for 87.24% of full estimated variance. Both estimates use the same returns, weights and 63-day window; their variance difference isolates the sample covariance contribution. Figure 28 retains the longer rolling history.

The existing tool lets users change dates, 63/126-day windows, and holdings. It also evaluates $\Sigma_\eta = (1-\eta)\,\mathrm{diag}(\widehat{\Sigma}) + \eta\widehat{\Sigma}$, with $0 \le \eta \le 1$. The interpolation preserves positive semidefiniteness and gives a conditional covariance scenario for the selected holdings. The interface exposes how estimated common movement changes the same portfolio's risk.

## 18.2 Strategy variants are not independent risk sources

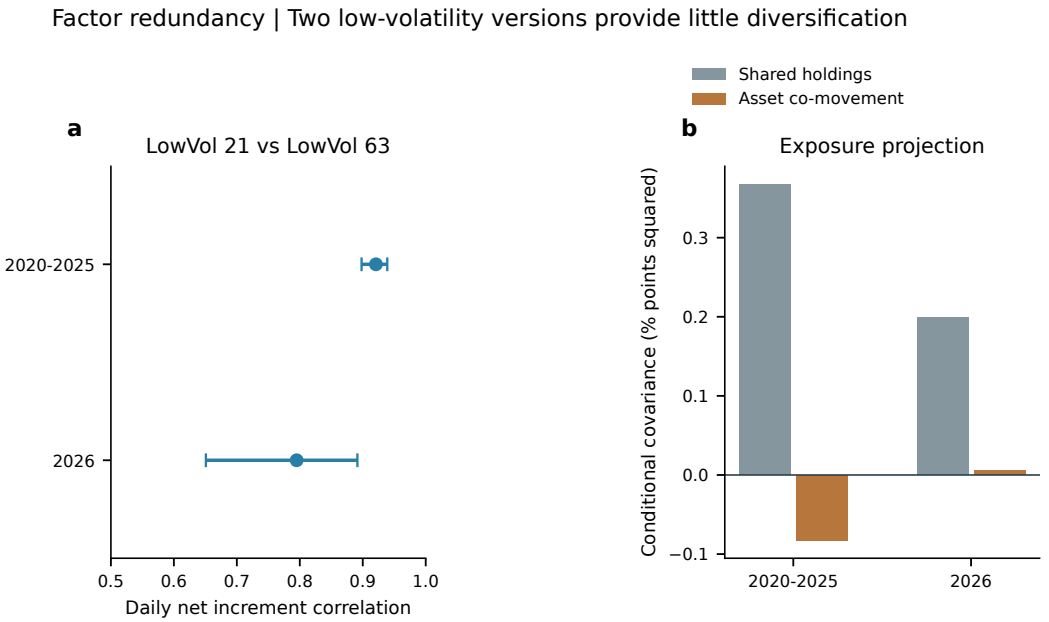
The six fixed proxies use momentum at 63/126 days, reversal at 5/21 days, and low volatility at 21/63 days. Dependence is assessed primarily through daily net P&L increments. The low-volatility pair has daily correlations 0.921/0.795 in the historical/2026 periods, versus cumulative-curve correlations around 0.982/0.985. Cumulative curves therefore cannot replace daily-risk analysis. Figure 29 separates shared exposures and cross-asset terms.

The default screening rule uses only 2011–2017 daily net increments and turnover. It ranks candidates by turnover and flags positive correlations $\ge 0.80$. Six candidates become five representatives,




a: Rolling 63-session estimates, descriptive. b: Per-asset trailing-scale standardisation.
499 within-year shifts; bars show central 95% of surrogates, not a sampling CI or causal test.
Data: Yahoo Finance daily ETF panel. Fixed research rules; no live trading claim.

**Figure 28.** Sector-portfolio risk decomposition and conditional shifts of cross-asset date alignment. Numerical risk uses the full covariance matrix, independently of the plotted edge threshold. Shift ranges describe the conditional alignment comparison. The fixed 2020 case is in risk_lab_v1/risk_snapshots.csv. [U02]




a: 20-session paired blocks, 4,999 resamples; post-analysis descriptive pair diagnostic.
b: Average past-covariance projection, no causal partition. Equal-risk effective count: 1.04 / 1.11.
Data: Yahoo Finance daily ETF panel. Fixed research rules; no live trading claim.

**Figure 29.** Daily net P&L dependence for the two low-volatility strategy variants, with cumulative-curve comparisons and exposure-based risk decomposition. Intervals use date blocks. For equal-risk strategies with daily P&L correlation $c_{12}$, $2/(1+c_{12})$ is a covariance-equivalent count. [U05]

reducing their count by 16.7%. Evaluation-period statistical reconstruction of the flagged strategy gives $R^2 = 83.9\%/64.1\%$. The rule was proposed after evaluation data had been inspected and

remains exploratory. Reconstruction $R^2$ is neither retained alpha nor an executable hedge return.

**Table 16.** Default screening from six strategies to five representatives.

| Period | Cumulative net P&L (%) | Annual P&L volatility (%) | Daily turnover (%) |
|---|---|---|---|
| 2020–2025 | -21.68→-20.77 | 4.35→4.33 | 7.46→8.67 |
| 2026 confirmation | -6.44→-5.57 | 3.84→3.99 | 8.03→9.05 |

Fixed reference capital with additive P&L; 5 bp one-way transaction cost and assumed 100 bp annual borrowing cost. Screening is post-analysis exploration, not a new blind test. Source: risk_lab_v1/subset_metrics.csv.

Table 16 shows higher combined turnover after removing one candidate, together with higher 2026 volatility. The useful workflow flags candidates, then checks combined holdings, costs, and exposures before deciding what to retain. The interface makes all 63 nonempty subsets available for comparing those consequences under a fixed evaluation period.

### 18.3 How the mechanism enters the tool

With additional assumptions of common loading $v$, independent noise covariance $D$, and fixed strategy exposures $W$,

$$\Sigma_Z = Kvv^\top + D, \quad K = q(\lambda/N)^2 \mathbb{E}S_2,$$
$$\Sigma_{\text{strategy}} = W\Sigma_Z W^\top. \tag{55}$$

This conditional factor projection connects a coordination-generated common mode to the covariance used by the risk tool. Empirical covariance can also reflect common news, holdings, and heteroskedasticity. The projection describes fixed exposures; daily strategy P&L is evaluated separately with its actual time-varying holdings.

The existing single-file tool has four modules: omitted risk, factor screening, allocation comparison, and physical mechanisms. Its mechanism module varies $\rho, q$, redraws graphs, and simulates actions while displaying cluster moments and model risk. The common-loading scenario is an illustrative model calculation; sector risk uses the observed return covariance. Tool settings, source metrics, and implementation checks accompany the interface.

Source tables and methods: `web/advanced/index.html`.

## 19 Discussion

### 19.1 The useful object is a weighted state-and-environment risk map

The study connects a tractable herding mechanism to dynamic risk without discarding its mathematical origin. The finite reference supplies exact moments and a structural identification boundary. Effective weights preserve the covariance mechanism when decisions are large or fragmented into many executions. The interdependent network adds fast/slow memory, adaptation and accommodation, and the calendar-time observation layer distinguishes decisions from recorded trades. The resulting object is a conditional risk map of effective dependence, trading weight and response environment.

The multi-factor continuation makes that map useful for a concrete question about quantitative trading. More strategy names need not diversify exposure; genuine input independence buffers risk, but in the tested grid does not offset fourfold effective order-weight expansion. Weak capacity can outweigh the diversity buffer. The separated channels also reject a blanket connection-to-instability claim: peer influence and actual-return feedback raise risk in the tested configuration, whereas fast–slow coupling is a net buffer. These are controlled responses of one stated economic mechanism, not estimates of the number or market share of quantitative institutions.

This interpretation connects aggregate-fluctuation models[3, 4] to evidence on crowded quantitative exposure[18] and the role of liquidity in transmission[19, 20]. The added contribution is their controlled connection through one weighted network, price law and observable interface, which

separates mechanisms often described together. It shows which interventions matter under common conditions and which intuitive links do not follow automatically.

## 19.2 Concentration, covariance and risk require matched timing

The observed factor panel qualifies the positive-loading input model: signed cross exposures hedge risk in some months and amplify it in others. The effective number of correlation modes is substantially below the six nominal labels, but neither label count nor a single positive common-share parameter describes all relevant structure. This makes signed exposure and its changing covariance a natural next representation within the original framework.

Both simulation and observed-return allocation tests also separate concentration from realized risk. The model establishes a concentration effect under its specific allocation rule, while its sixteen-hour risk increment is unresolved. The observed-return performance rule is more concentrated than fixed allocation in the primary block analysis, but its score-permuted control is more concentrated still, and the six-month dependence sensitivity weakens the fixed-allocation result. Terminal model weights govern a future bar; empirical held weights precede their measured next month. Keeping those clocks explicit prevents a concentration statistic from being mistaken for the cause of risk already accumulated. A future capital-flow model needs actual rebalance orders, financing and endogenous price response before concentration alone can support a destabilization claim.

## 19.3 A mechanism effect is not automatically new forecast information

The precision study resolves genuine heterogeneity in covariance and feedback effects after reducing Monte Carlo noise. The information study finds a strong joint contribution from hidden activity, size, accommodation and weight concentration. Nevertheless, an effect estimate can repeat information already summarized by activity and volatility. Exact local-noise cancellation and direct-moment controls show how a mechanism-shaped representation may encode an ordinary return moment. The inverse target is therefore a distribution over effective states that change future risk, rather than a uniquely recovered trader count or a coordination scalar selected for interpretability alone.

Retaining state and parameter uncertainty improves distributional prediction over legal point-state propagation under a finite source-mixture law; CDF-shaped calibration allocates interval width more efficiently. Their direct transfer to Bitcoin is insufficiently calibrated, and the earlier internal HAR augmentation point advantage does not recur on the frozen 11-date replay. The later Bitcoin closure's 5.79% internal point gain remains statistically unresolved and is not uniquely attributable to coordination. The new factor-return comparison likewise retains the simpler HAR-Portfolio, which has the best point score. Taken together, the results support executable explanatory and uncertainty tools, while additional market forecast value and total information savings remain open empirical tests.

## 19.4 Applications and the next discriminating observation

The portfolio, strategy-overlap and historical replay tools expose common risk, distinguish observed quantities from model estimates and allow conditional inspection. Explore adds the environment and strategy-ecology maps: input dependence, effective scale, accommodation and allocation can be varied with paired uncertainty and explicit horizons. Observed factor diagnostics occupy a separate layer. This gives users an explanation of why similar strategies may be risky in one receiving environment but buffered in another, rather than an unvalidated live trading signal.

The next discriminating observation is a low-cost measure of signed exposure or price response that helps separate common pressure from the environment receiving it. A fixed candidate must be supplied to every statistical reference and tested on genuinely new dates or another asset, with parameter-calibration uncertainty and the remaining activity/persistence mismatch carried forward. Real strategy holdings, flows or execution-response data would also distinguish endogenous capital concentration from a computed allocation rule. Price discovery, execution costs and welfare would be required to infer a regulatory trade-off or an optimal limit on quantitative trading; the current risk scenarios supply no such threshold.

Quantum models and quantum machine learning remain possible extensions for collective dynamics and inference. A useful first task would compare observation requirements, error and computational cost on the identifiable known-truth state problem. A common encoding cannot separate classical observation laws that are already identical; the choice depends on a specific inferential or computational benefit[14], not an assumed advantage or implementation timetable.

## 20 Conclusion

Aggregate observations, effective weighted dependence and dynamic price risk form one continuous research problem. Exact herding mathematics defines the starting inverse and its limits; execution diagnostics, fast–slow dynamics and calendar-time response extend that mechanism toward market observations. The completed scenario experiments show that genuine factor diversity buffers risk, while effective scale, accommodation, peer influence and feedback govern the size and even the sign of transmission channels. Slow allocation can concentrate exposure without establishing a risk increment at the measured horizon. Observed factor returns add a signed-covariance check and show why nominal diversity and concentration require direct risk tests. Source-law uncertainty and shape calibration provide useful distributional methods, while Bitcoin transfer, direct-moment controls and matched-history HAR comparisons retain the distinction between explanation and additional prediction. The established contribution is an executable, quantitative explanation of collective-risk transmission and the observable evidence needed to assess it. Reliable incremental market prediction and institution-level quantitative-trading attribution remain unresolved.

## A Model context: adaptation, bubbles and crashes

The model families in Table 17 address complementary mechanisms. Cont–Bouchaud links random coordination and shared cluster actions to aggregate fluctuations[3]. Our finite implementation specifies both volume observation and unit-order execution. Lux–Marchesi adds switching between opinions and trader types, producing endogenous feedback and temporal structure[4]. The persistent-state experiment in Section 6 isolates temporal activity and response while retaining exogenous state dynamics.

Johansen–Ledoit–Sornette (JLS) relates reinforcing imitation to conditional bubble growth and stochastic crash risk[5]. Its simplified setup is

$$dp/p = \mu(t)\,dt - \kappa\,dj, \quad \mathbb{E}[dj] = h(t)\,dt, \quad \mu(t) = \kappa h(t). \tag{56}$$

The hazard $h(t)$ links growth conditional on survival to compensation for jump risk. The study discusses pre-crash patterns for Wall Street in 1929, 1962 and 1987, and Hong Kong in 1997. These published retrospective cases illustrate the bubble mechanism. The endpoint of conditional bubble growth and the random crash time are distinct model quantities.

**Table 17.** Mechanisms represented by three financial market models.

| Model | Mechanism | Connection to the present analysis |
|---|---|---|
| Cont–Bouchaud[3] | Shared actions in random groups | Finite ER model, price–volume moments, inversion and compatibility tests |
| Lux–Marchesi[4] | Opinion and trader-type switching with profit feedback | Adaptive comparison for the temporal-state experiments |
| JLS[5] | Reinforcing imitation, hazard and conditional growth | Literature context for bubbles, crashes and criticality |

## B Complete tail and scaling objects

Cluster number density, node-weighted cluster size, and return distributions are different observables. Figures 30 and 31 retain cluster evidence and separate population size from temporal aggregation.

Independent-model second-order aggregation differs from market memory.

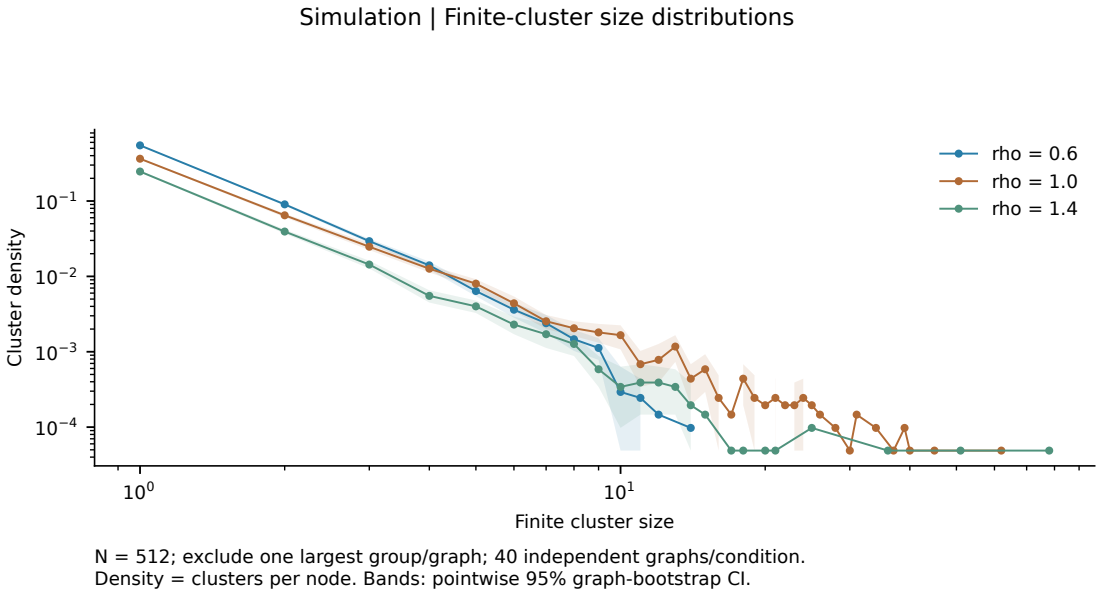



**Figure 30.** Finite-cluster number density at the specified population size. Pointwise bands resample graphs. Uniform-node sampling would weight each cluster by its size; raw cluster sizes and all population-size source tables are retained. [F04]

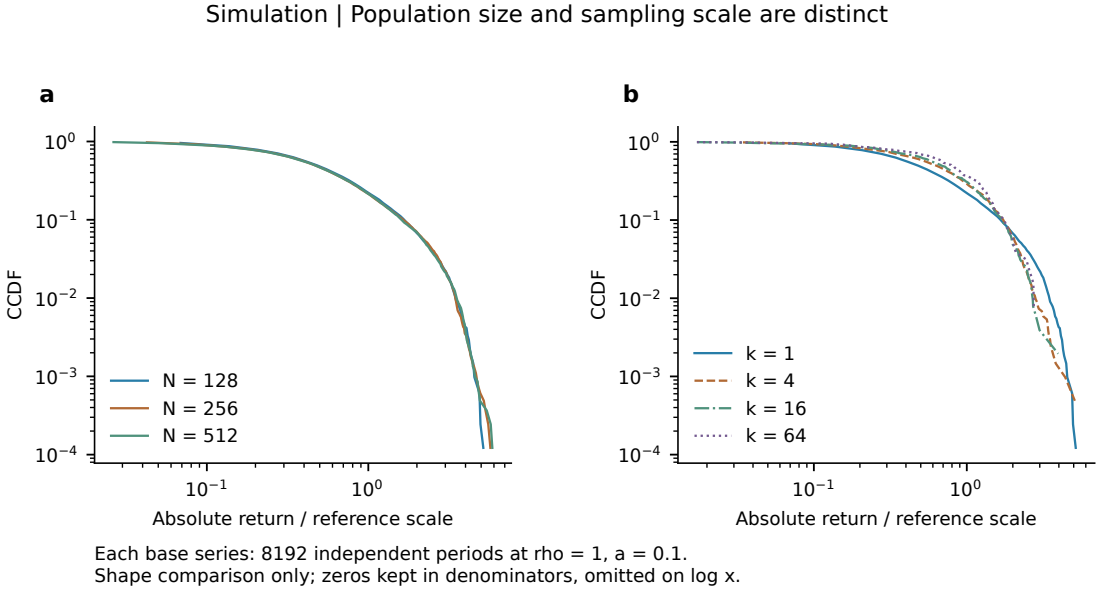



**Figure 31.** Separate comparisons of population size and temporal aggregation. Return standardization permits shape comparison; temporal blocks are nonoverlapping. The source tables retain all conditions. [F06]

Figure 32 adds the calibration-scaled market/model tail comparison to the joint-observation and memory diagnostics. All 28 market CSVs and original reproduction records are retained. Source maps distinguish model seeds from observable inputs.

## C Exact checks and dependence controls

Theory files contain proofs of Eqs. (2)–(3), monotone identification, and chain/star and mixture equivalence. Rational enumeration and exact certificates check implementations. The original 480 theory checks and ten-study refit records are included. Small-graph enumeration and rational certificates provide independent computational checks of the stated identities. The marginal-

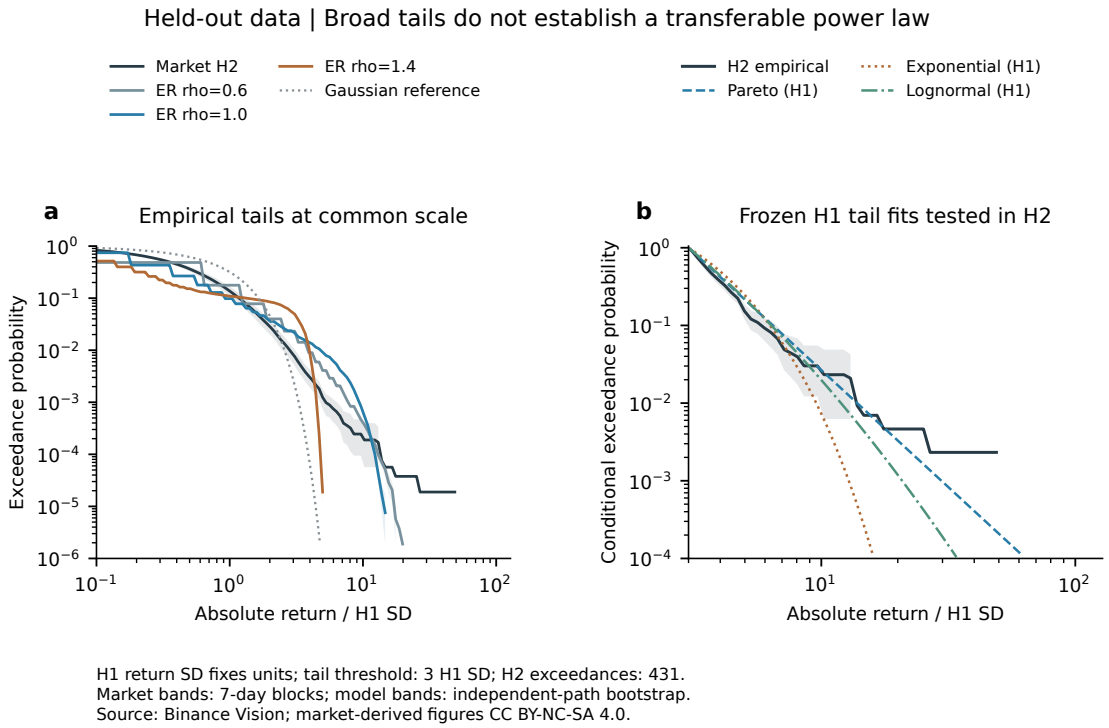



**Figure 32.** H1-scaled H2 market return CCDFs, conditional tails and continuous alternatives. Zero exceedances remain in the source data and are omitted from logarithmic display. Binance Vision, CC BY-NC-SA 4.0. [R02]

preserving permutation comparison D03 isolates same-period coupling: shuffling volume preserves both empirical marginal distributions while changing their joint dependence.

# D Supplementary forecasts and external evaluations

For a positive realized variance $y$ and positive prediction $\mu$, the quasi-likelihood loss (QLIKE) is

$$\ell_{\mathrm{QLIKE}}(y, \mu) = y/\mu - \log(y/\mu) - 1. \tag{57}$$

QLIKE is a secondary loss in the original log-RV benchmark and the primary loss in the subsequent state-forecast study. The latter's Gamma specifications use a log-linked positive mean and separate penalties on return and trading-information channels. Off denotes the matched return-only model with the trading-information channel fixed to zero. Feature and fitting definitions are in `docs/AB/state_forecast_v2_feature_spec.md`.

**Table 18.** Supplementary forecast losses, without replacing the primary metric.

| Model | QLIKE | Volatility MAE (percentage points) |
|---|---|---|
| HAR | 0.234214 | 0.440987 |
| HAR+All | 0.233933 | 0.433735 |

Volatility is $100\sqrt{RV}$. Exponentiating a log prediction need not yield a conditional arithmetic mean.

Table 18 retains supplementary losses. Monthly point gains over HAR are 2.90%, 5.41%, 4.99%, −1.12%, −1.71%, and 5.15%. A post-analysis comparison against HAR+Tail gives a 1.535% point gain, again with an interval crossing zero. The final evaluation contains eight high-volatility events, giving a small event-specific assessment sample.

The subsequent state study adds an external ETH evaluation and 25 new dates for each of BTC and ETH. BTC's new-time primary QLIKE gain is 2.476%, with approximate 95% interval $[-0.407, 4.815]$%. Validation selects Off for the ETH trading channel. Complex Gamma specifications do not consistently beat simple HAR, as retained in Fig. 33. The BTC and ETH new-time evaluations cover the same 25 calendar dates.

On the same short BTC period, supplementary log-MSE improves by 3.170% against matched return-only Gamma. Its approximate 95% interval is $[1.236, 4.733]$%, with approximate Holm $p = 0.0276$ across that three-test supplementary family. This contrast assesses the trading-information channel within the Gamma specification. The primary assessment uses QLIKE and retains simple HAR among the references. Twenty-five dates provide only about 3.6 seven-day blocks, requiring a strong dependence approximation.

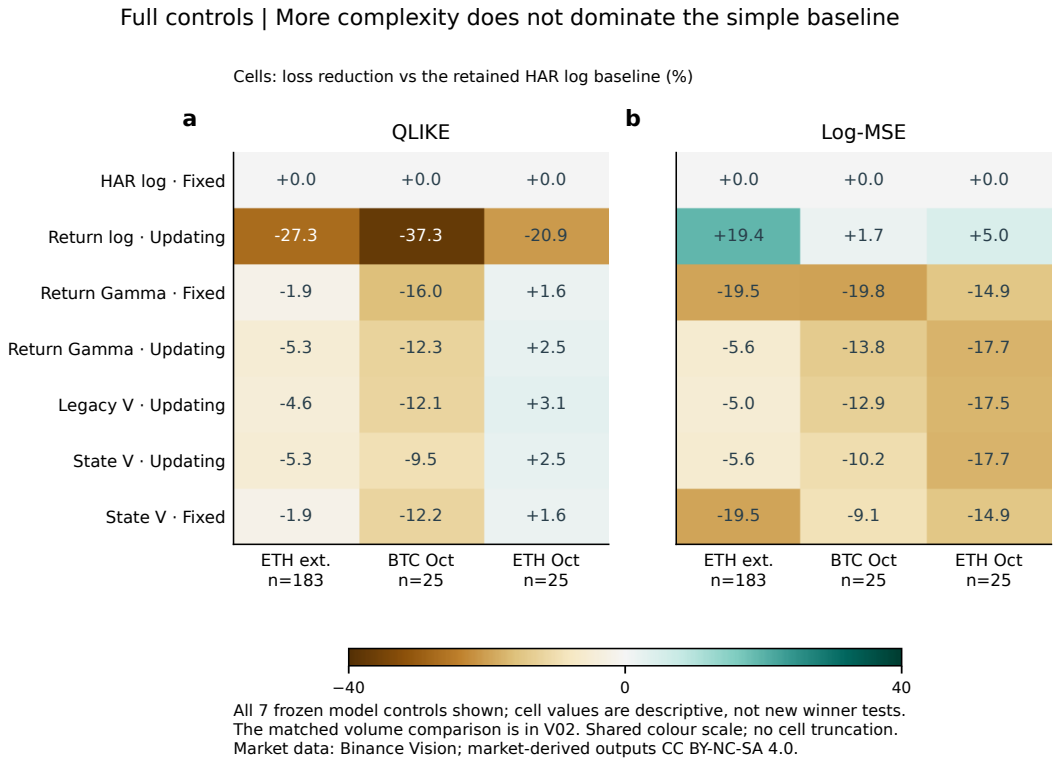



**Figure 33.** State/price–volume forecasts with simple HAR and matched controls. Previously inspected BTC periods are development information; external-asset and new-time evaluations are reported separately. Off denotes a validation-selected return-only fallback. Binance Vision, CC BY-NC-SA 4.0. [V03]

Covariance-matched tail tests hold information and covariance fixed, comparing nine methods on 38,070 rows from nine external stocks. Historical tail improvement is 0.557%, with Holm $p = 0.0772$; the 2026 point change is $-1.797$%. There are 32 historical co-crash events and zero in 2026, restricting the latter evaluation to false alarms. Figure 34 and the complete controls do not establish stable market increment. The external-stock evaluation tests transfer across assets over an already elapsed historical period.

# E Stopped capacity extension and volume trade-offs

Six strategies, nine ETFs, 190 monthly snapshots, four allocations, and three depth levels produce 2,280 conditional states. Volume-only, one-round impact, and multiround-feedback capacity bounds coincide in every state, giving 0% extra reduction in Fig. 35. The maximum initial loss, 1.79158%, is below the fixed 2% trigger. A certificate also verifies nonactivation at smaller scales. The fixed trigger is inactive throughout this design, which makes this branch a certificate of nonactivation under the tested self-impact scenario.

A conventional volume proxy still exposes a useful trade-off. Among 189 Equal-allocation historical replays with three complete future days, mean-volume sizing exceeds the participation proxy 142 times, or 75.13%. Using the past-20-day 20th-percentile dollar volume reduces this to 62 cases, or 32.80%, while mean deployed capital falls to 72.90%. These historical replays compare capital utilization with subsequent participation-proxy exceedances for a conventional volume rule, separately from the inactive feedback scenario.

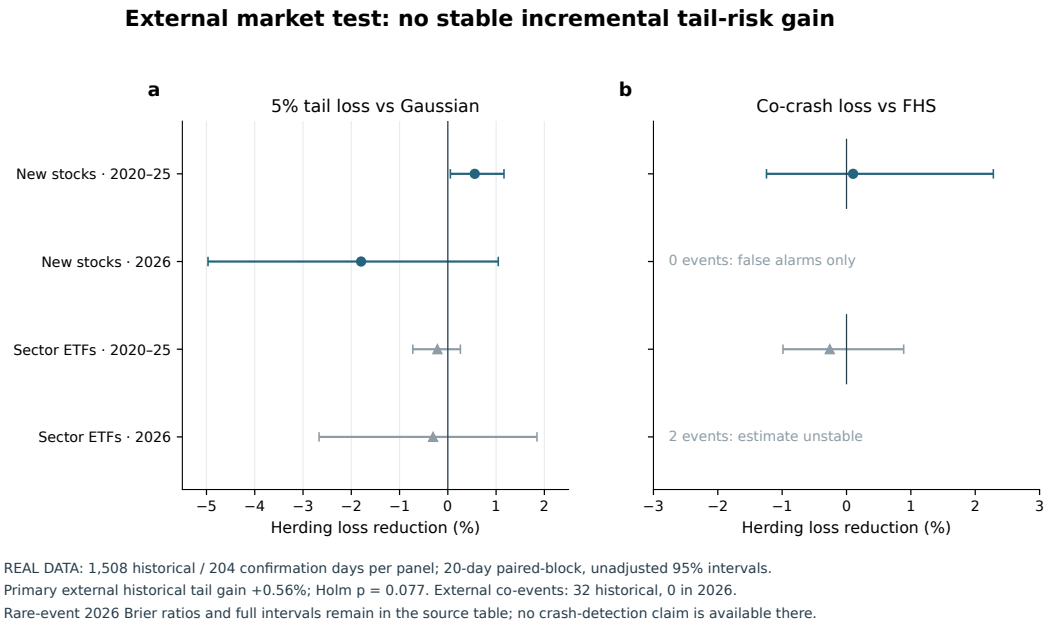



**Figure 34.** Tail and joint-downside comparisons at matched information and covariance. Statistical references, shrinkage, integration checks and criticality ablations are retained. No co-crash events occur in the 2026 sample, so that period evaluates false alarms. [T03]

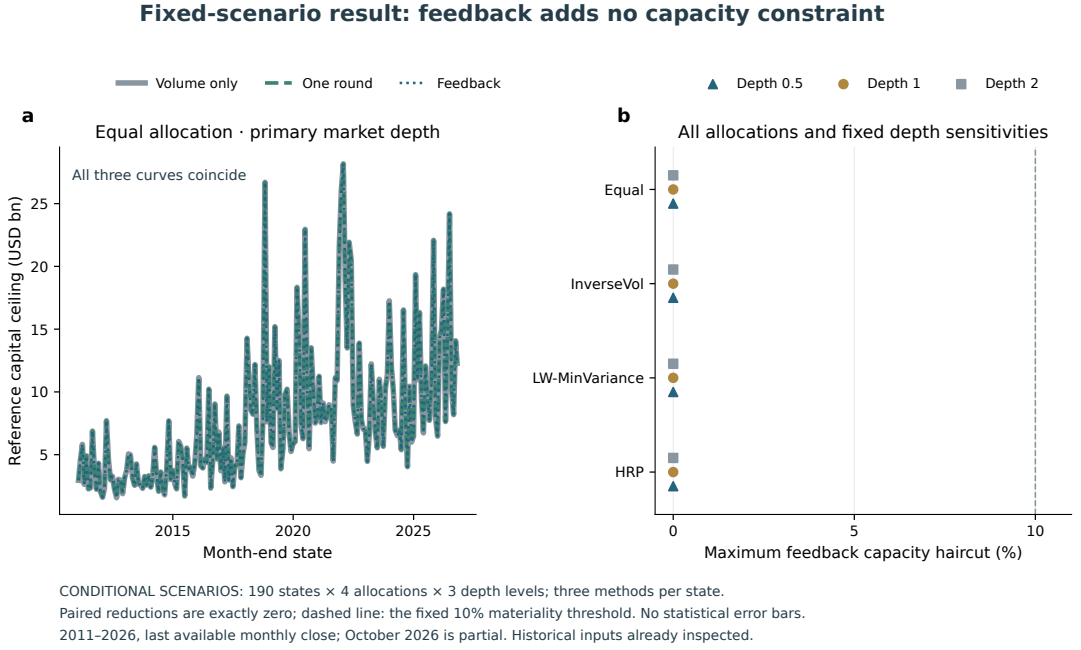



**Figure 35.** Coincident capacity bounds in the fixed account self-impact scenario. Historical prices and volumes supply conditional inputs; the feedback rule is specified rather than calibrated from execution data. All 2,280 states remain in the source record. [C01]

The package includes full code, nine capacity CSVs, cash and monotonicity checks, C01–C03 plotting code, and the original certificate.

# F Supplementary strategy-allocation paths

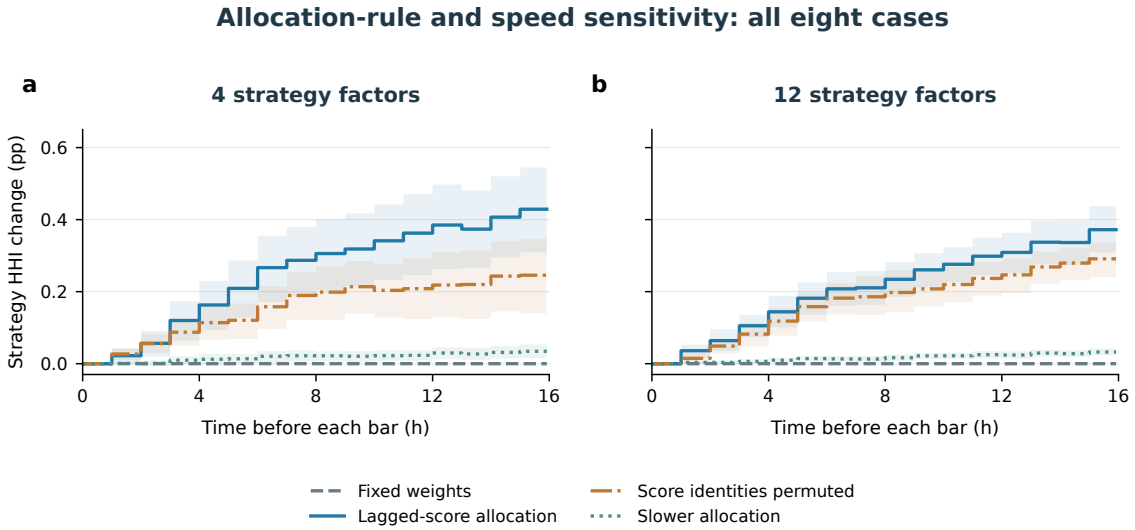



**Figure 36.** Pre-bar exposure concentration, allocation paths and slower-learning sensitivity in the same 192-bar model experiment. The terminal post-update HHI is kept separate from within-window pre-bar HHI. Performance, fixed, score-identity-permuted and slower-learning allocations use causal completed scores, positive normalized shares and refreshed network influence weights. Source: strategy-ecology allocation study.

# G Research record and reproduction

## G.1 Finding the evidence

The public repository and research website link the shared implementation in `src/econophysics/`, configurations, numerical tables in `results/tables/`, source manifests and three executable notebooks. Bracketed figure identifiers retain the original F/D/R/P/U/V/T/C/J mapping; named continuation figures document the weighted dynamic extension. Table 19 lists the reproduction entry points.

**Table 19.** Reproduction profiles from the repository root. Each run writes new outputs.

| Profile | Operation | Evidence |
|---|---|---|
| quick | Exact theory, model and source-table checks | Standard-library mathematics and numerical identities |
| theory | Exact identities and certificates | Calculations under the stated model assumptions |
| notebooks | Execute all three research notebooks | Model moments, inversion, market tests and forecast comparisons |
| site | Check local resources and the browser engine | Website links and agreement with Python fixtures |
| forecast | Download documented archives into a fresh directory and rebuild the fixed experiment | All 183 historical test dates and 16 saved CSV tables |
| Extended laboratory | Rebuild retained experiment pages | Risk, strategy overlap, allocation and coordination diagnostics |

### G.2 Dynamic continuation evidence

The public package in `studies/collective_risk_continuation_v3/` contains numerical source tables, original hashes, protocols, independent arithmetic checks and the Explore data export. It distinguishes source-law simulations, the conditional-ensemble design and market-transport diagnostics. Primary ensemble splits are 2,048 bank, 32 tuning, 64 calibration and 128 evaluation trajectories. The earlier particle-model calibration is a separate design; its scores are not pooled with this evaluation.

### G.3 State–environment diagnosis and interactive evidence

The curated release in `studies/state_environment_v9/` retains fixed numerical tables, uncertainty results, verification records and provenance. It contains the 256-origin precision and information studies, the complete 29-cell response grid, paired environmental contrasts, all internal market scores and the post-run direct-moment controls. Figures prefixed `v9_` reproduce those source results. The independent validation record distinguishes arithmetic/implementation checks from scientific admission criteria.

The interactive entry is Explore: Risk environment, with the dedicated risk-environment page. Scenario estimates, latent capacity and intervention uncertainty remain visibly separate from observed Bitcoin outcomes. The downloadable state–environment evidence bundle supplies the fixed release tables. Earlier network, inversion, forecasting and portfolio evidence is retained alongside this continuation.

### G.4 Strategy-ecology continuation and source-calibration results

The strategy-ecology release preserves the original 48-unit decision law and provides the multifactor mathematical bridge, complete factor/scale/allocation tables, paired contrasts and path-accounting checks. Figures prefixed `v10_` reproduce its vector source figures without modification. The experiment records 56 short-horizon cells and eight slow-allocation cells, with their separate origins, horizons and testing families. Complete paths and source hashes distinguish implementation verification from statistical evidence. The calibrated-state source study supplies the equal-192-future uncertainty comparison, source-CDF interval evaluation, market transfer and frozen 11-date replay. Those model-source and market cohorts are retained separately from the later state–environment study.

### G.5 Environment and source records

Forecast reconstruction uses Python 3.14.4 and the versions in `requirements-study.txt`; plotting and notebook dependencies are listed separately. A fresh reconstruction reproduces all 16 saved forecast tables byte for byte. Notebooks retain executed outputs, and the quick and website checks pass on Python 3.11 and 3.14.

Binance Vision-derived results retain CC BY-NC-SA 4.0 attribution. Full supplier price histories are acquired through the source manifests. Code, data and third-party asset terms are documented in `LICENSE.md` and `DATA_LICENSES.md`. Historical experimental digests remain in the source record.

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